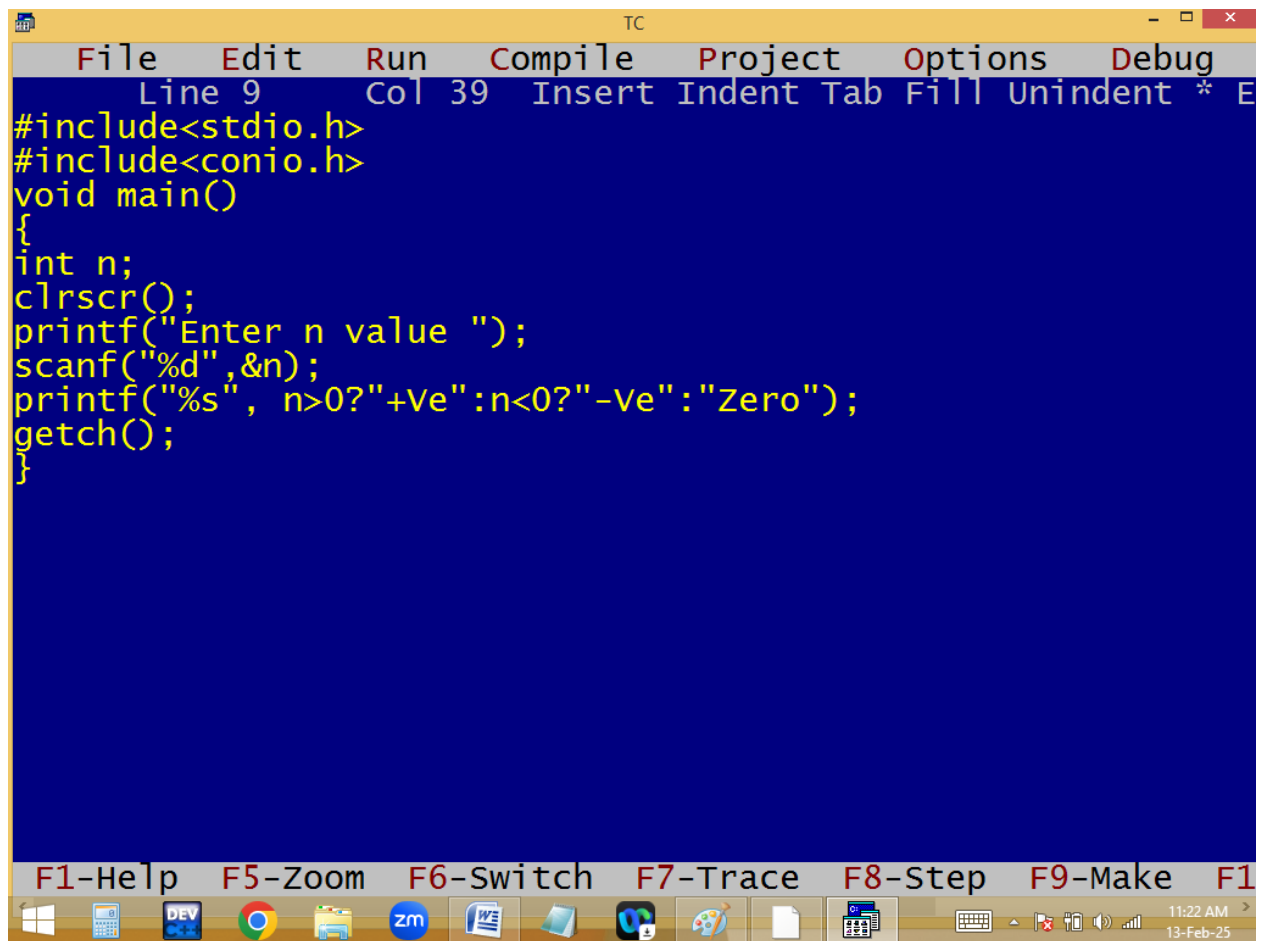


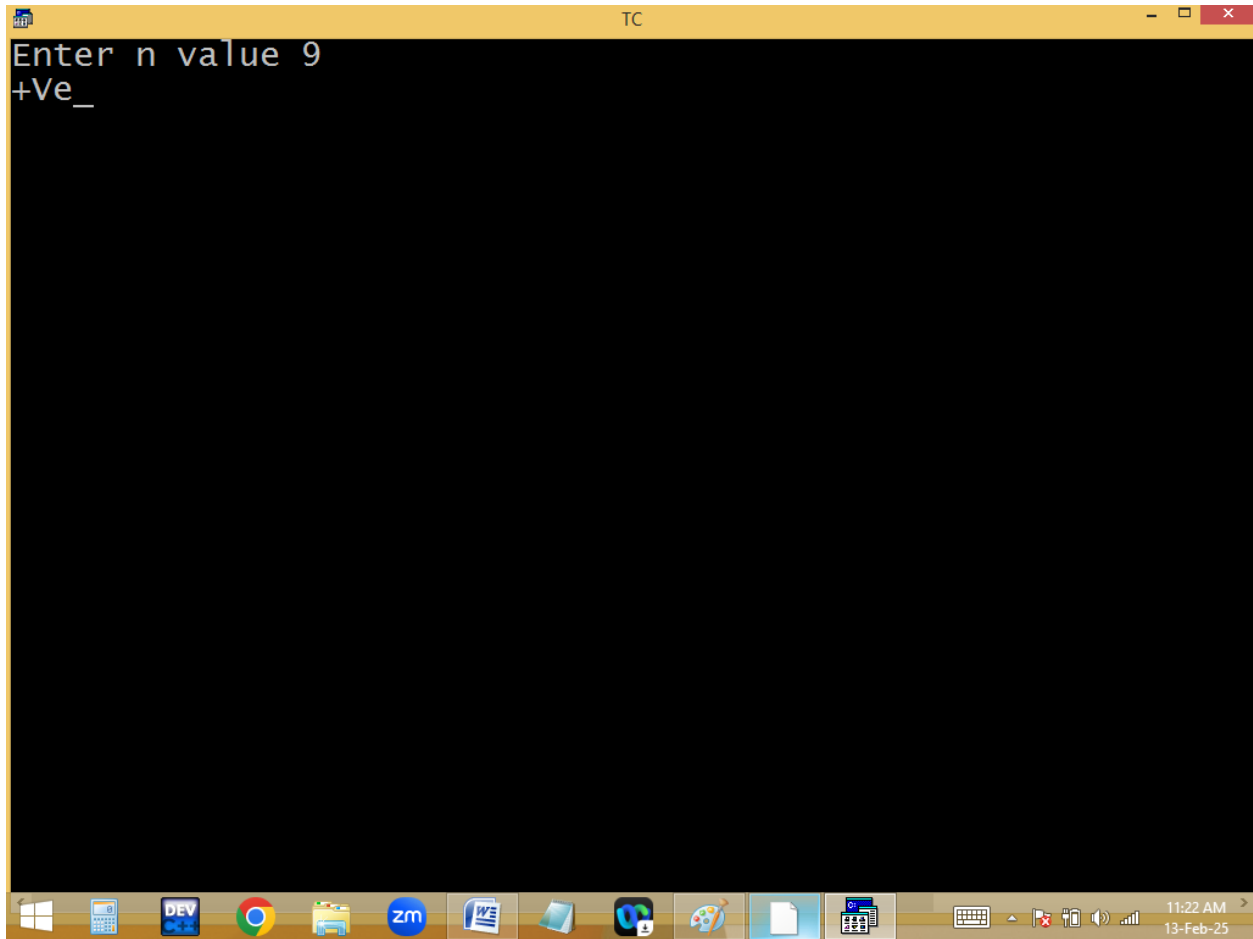
Finding +/-/0 using ternary operator?

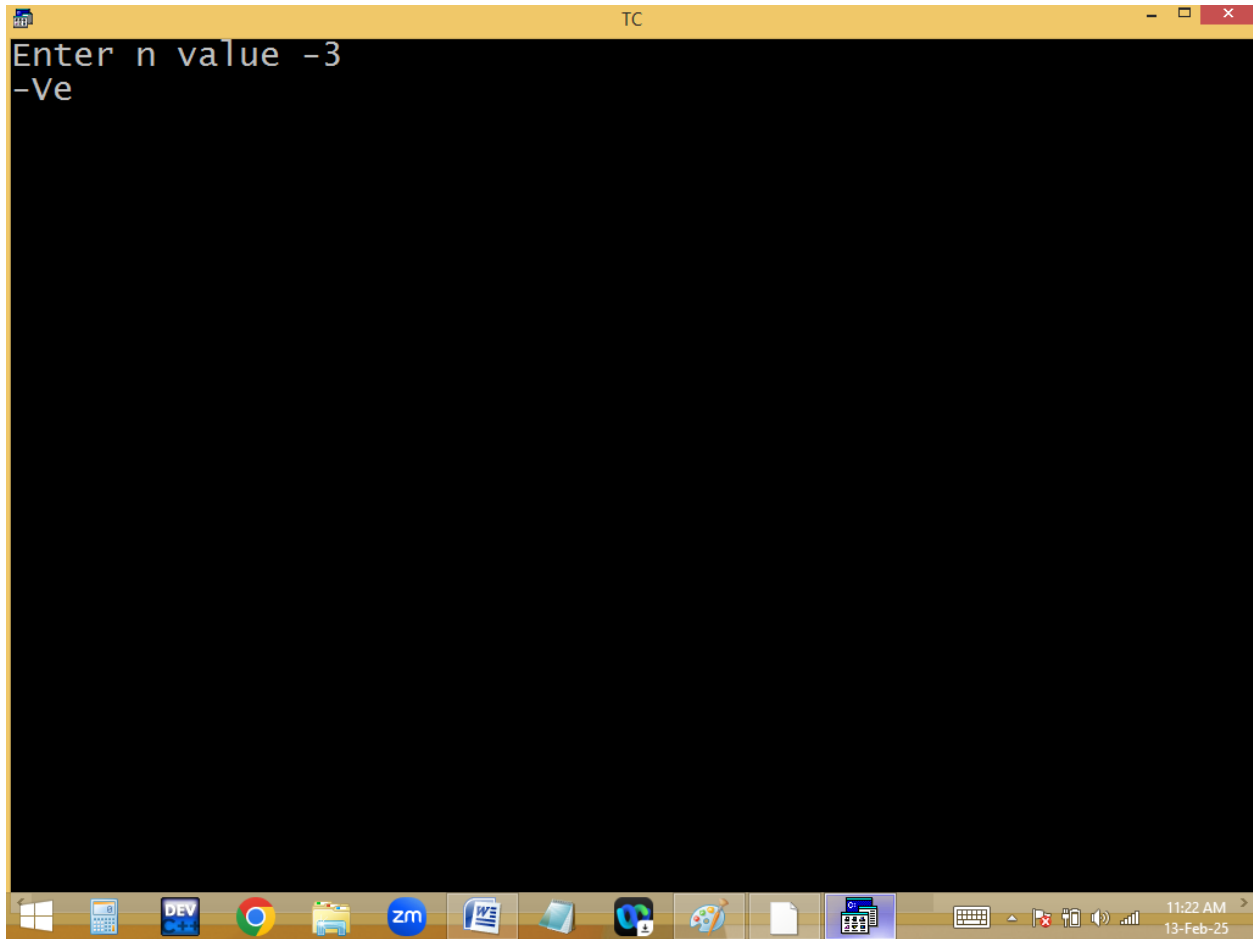


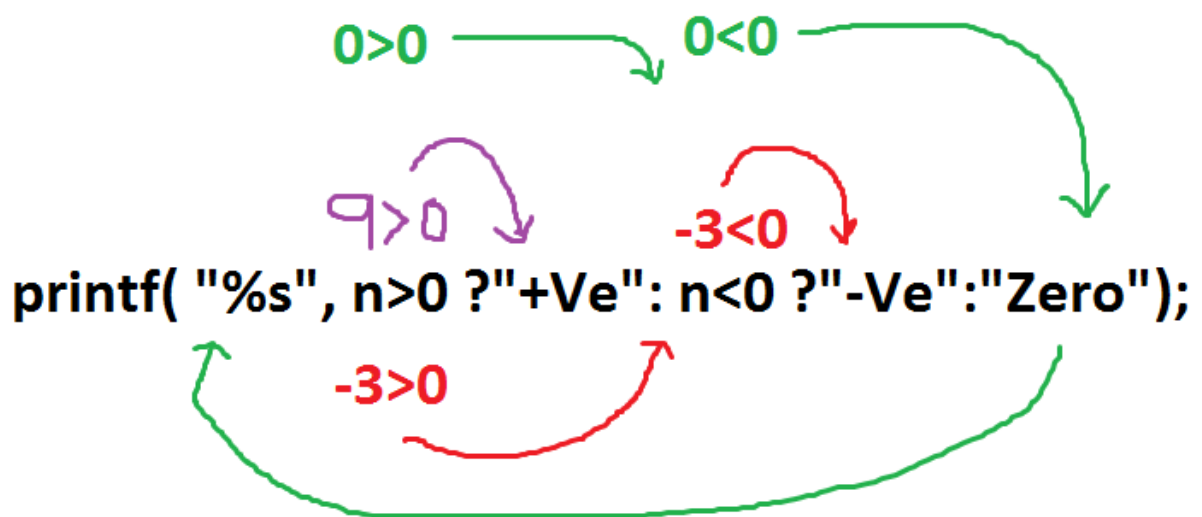
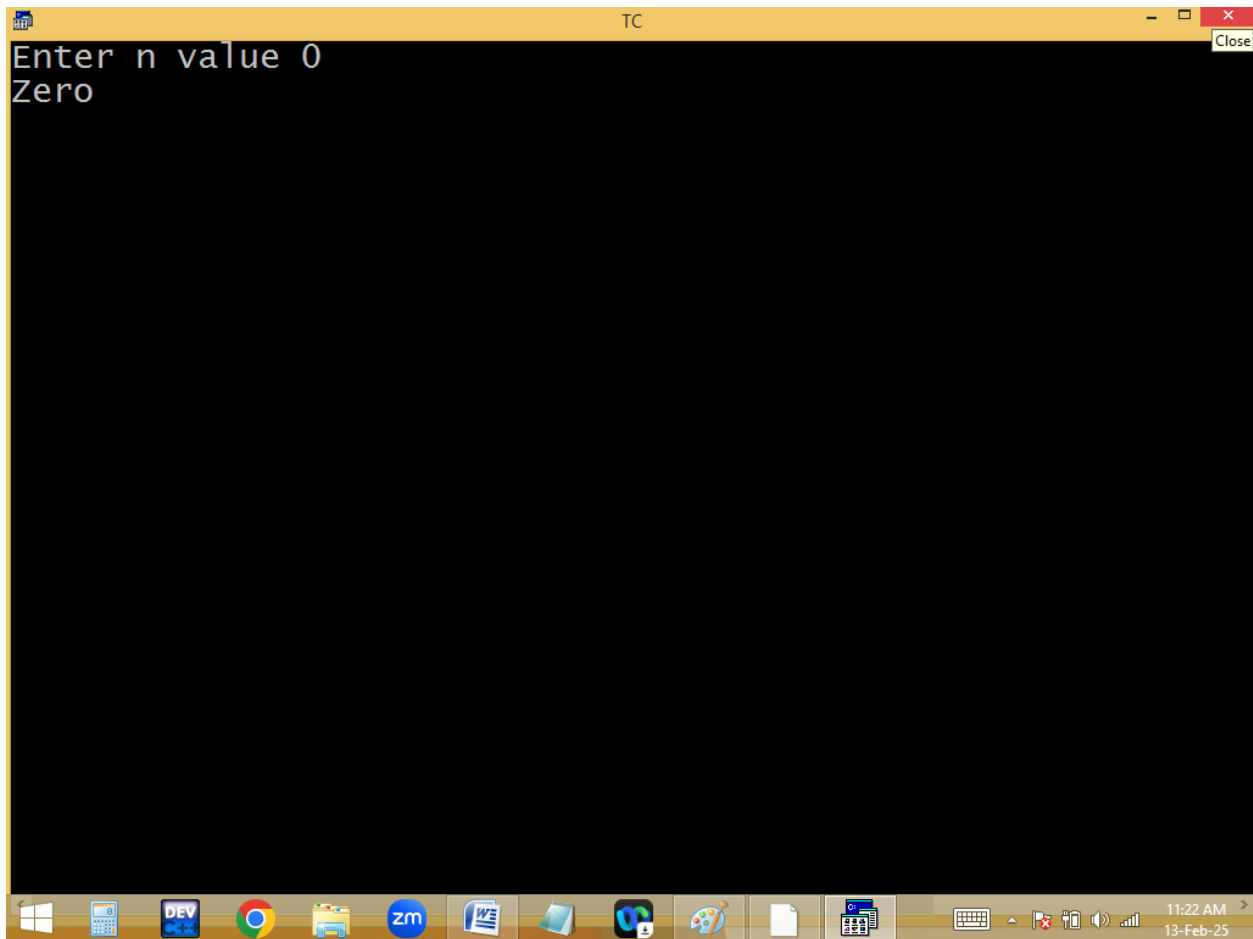
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 9", "Col 39", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a blue background and contains the following C code in yellow text:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value ");
scanf("%d",&n);
printf("%s", n>0?"+ve":n<0?"-ve":"Zero");
getch();
}
```

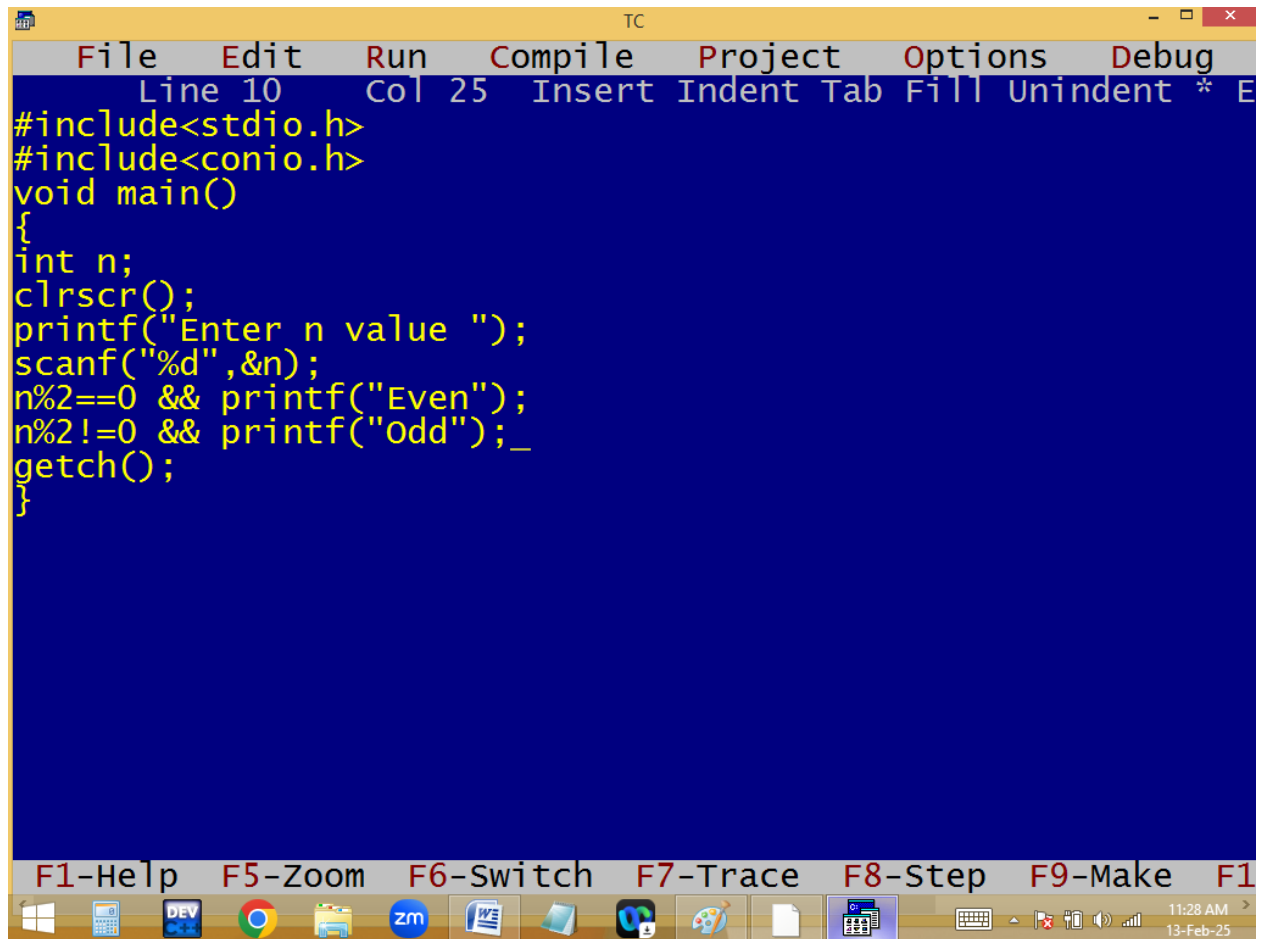
At the bottom of the window, there is a toolbar with icons for various functions, each with a corresponding F-key shortcut: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar is visible at the very bottom, showing the Start button, taskbar icons for Windows Explorer, Dev-C++, Google Chrome, File Explorer, Zoom, Notepad, and others, along with the system clock showing 11:22 AM on 13-Feb-25.







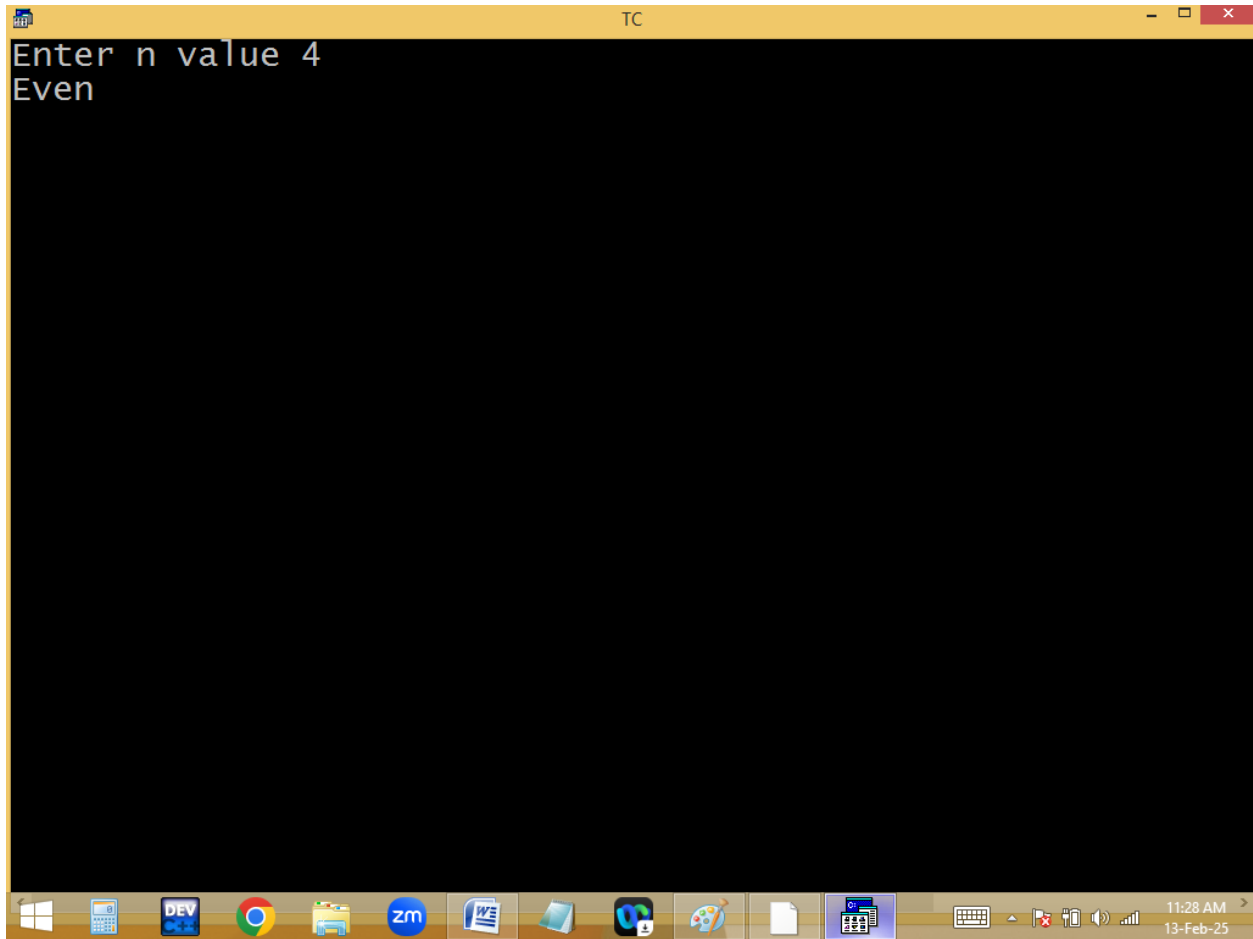
Finding even/odd without using if..else/ternary op?

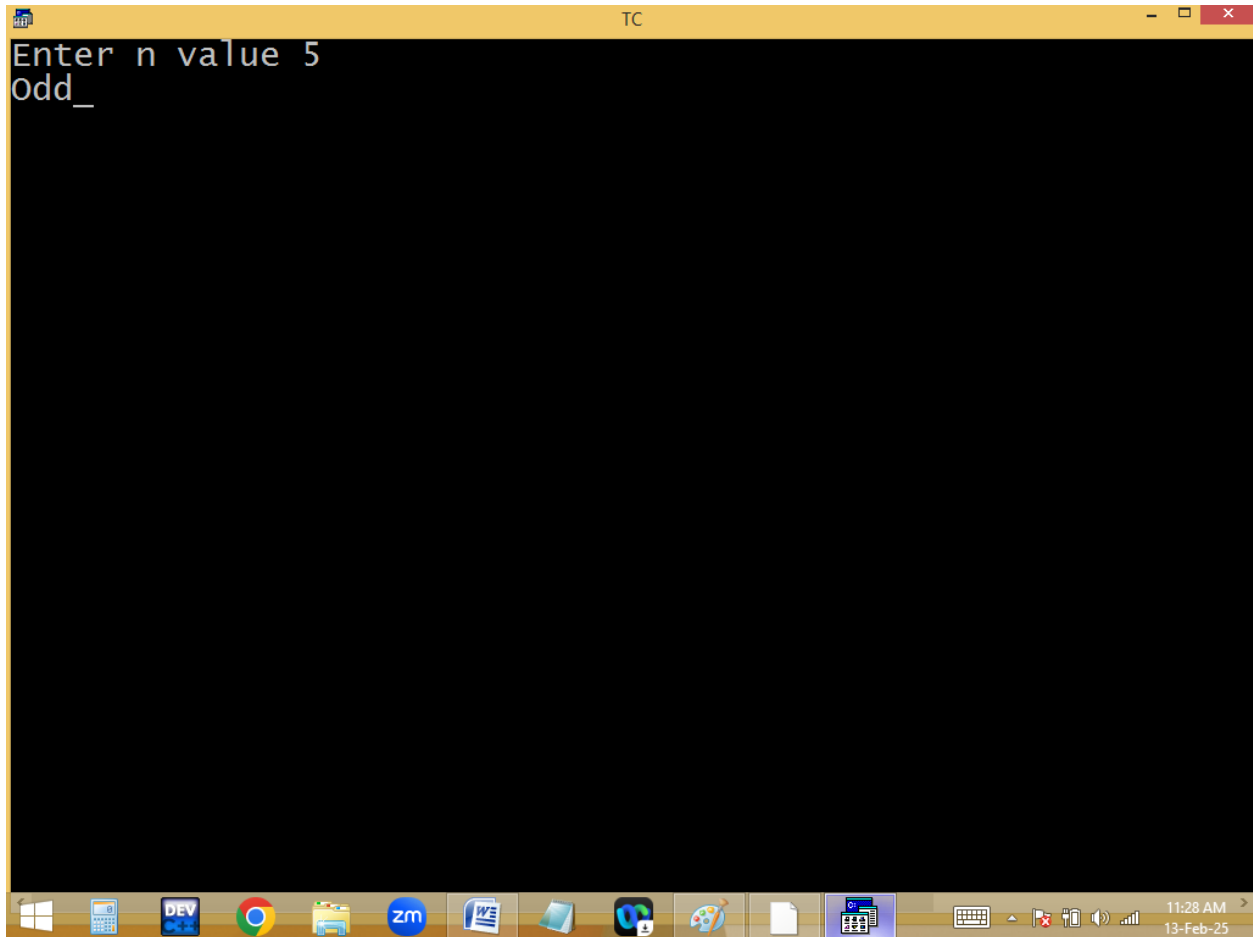


The image shows a screenshot of the Turbo C++ (TC) integrated development environment. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar indicates "Line 10", "Col 25", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and a multiplier "* E". The main editing area has a dark blue background with yellow text. The code is a C program that checks if a number is even or odd. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value ");
scanf("%d",&n);
n%2==0 && printf("Even");
n%2!=0 && printf("Odd");_
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions. Below the toolbar, a status bar displays function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The system tray at the bottom right shows the time "11:28 AM" and the date "13-Feb-25".





$4\%2=0==0$

$n\%2==0 \ \&\& \ \text{printf}(\text{"Even"});$

$4\%2=0!=0$

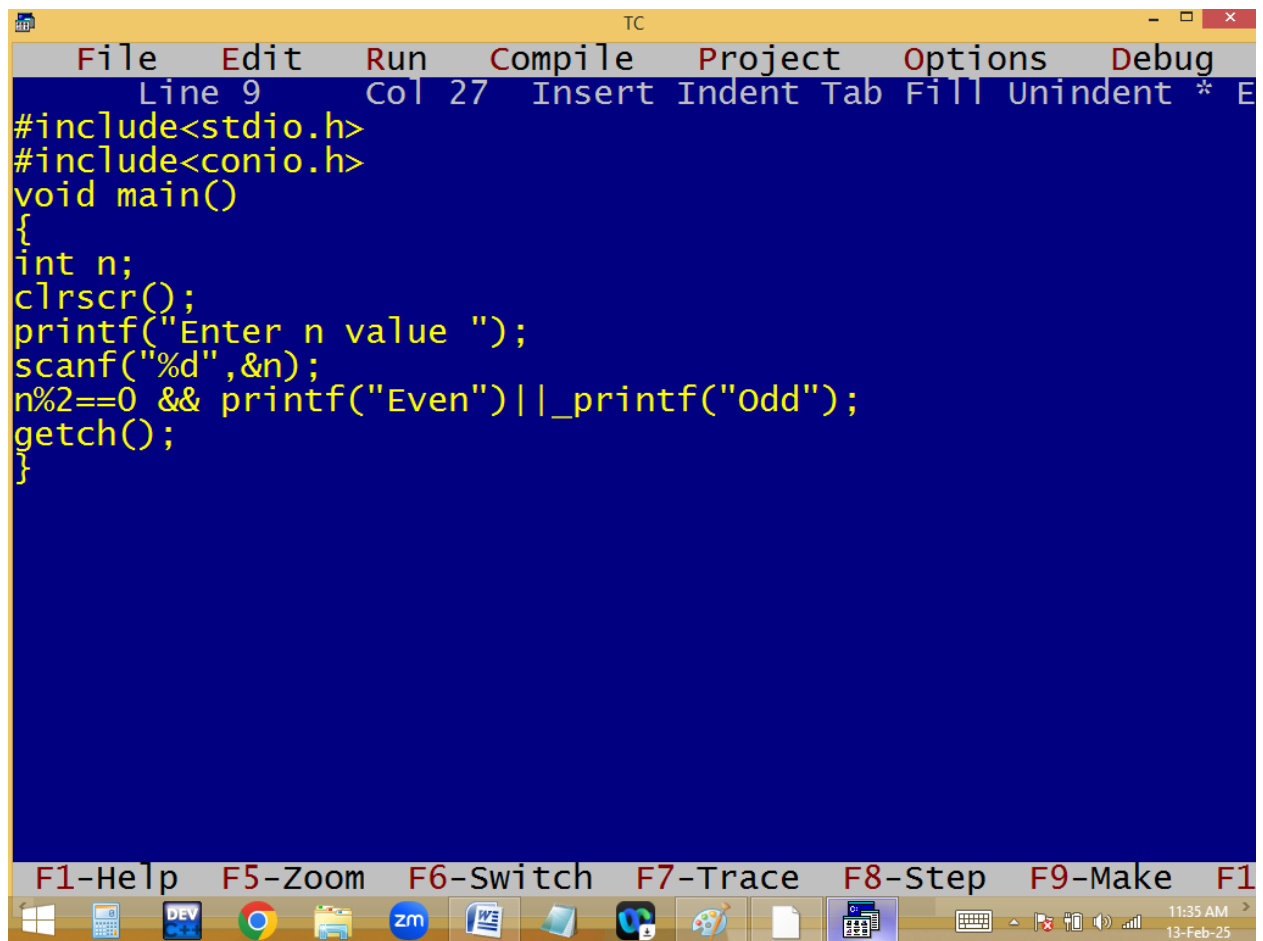
$\times$
 $n\%2!=0 \ \&\& \ \text{printf}(\text{"Odd"});$

$5\%2=1==0 \implies \text{False}$ $\times$

$n\%2==0 \ \&\& \ \text{printf}(\text{"Even"});$

$n\%2!=0 \ \&\& \ \text{printf}(\text{"Odd"});$

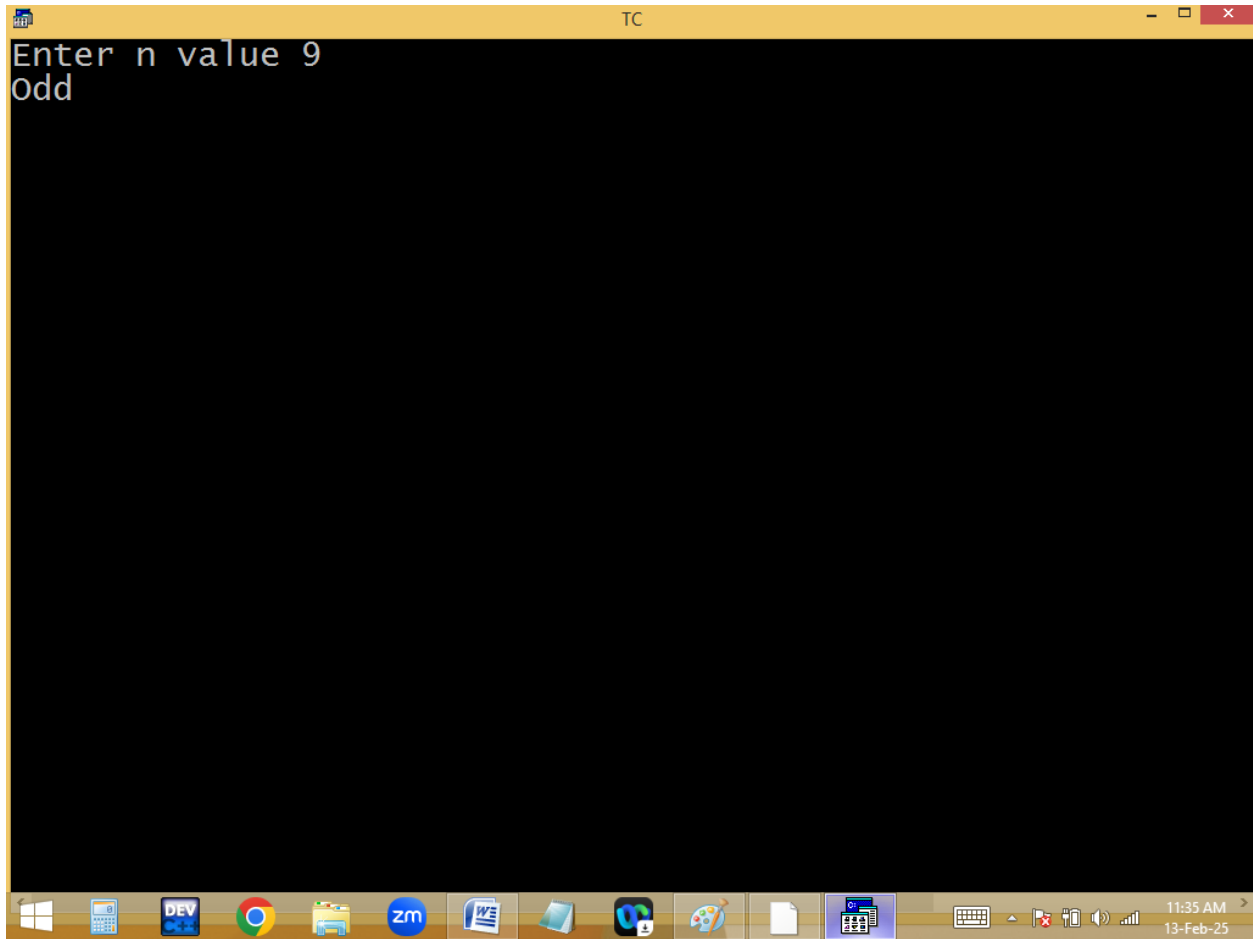
$5\%2=1!=0 \implies \text{true}$ $\curvearrowright$

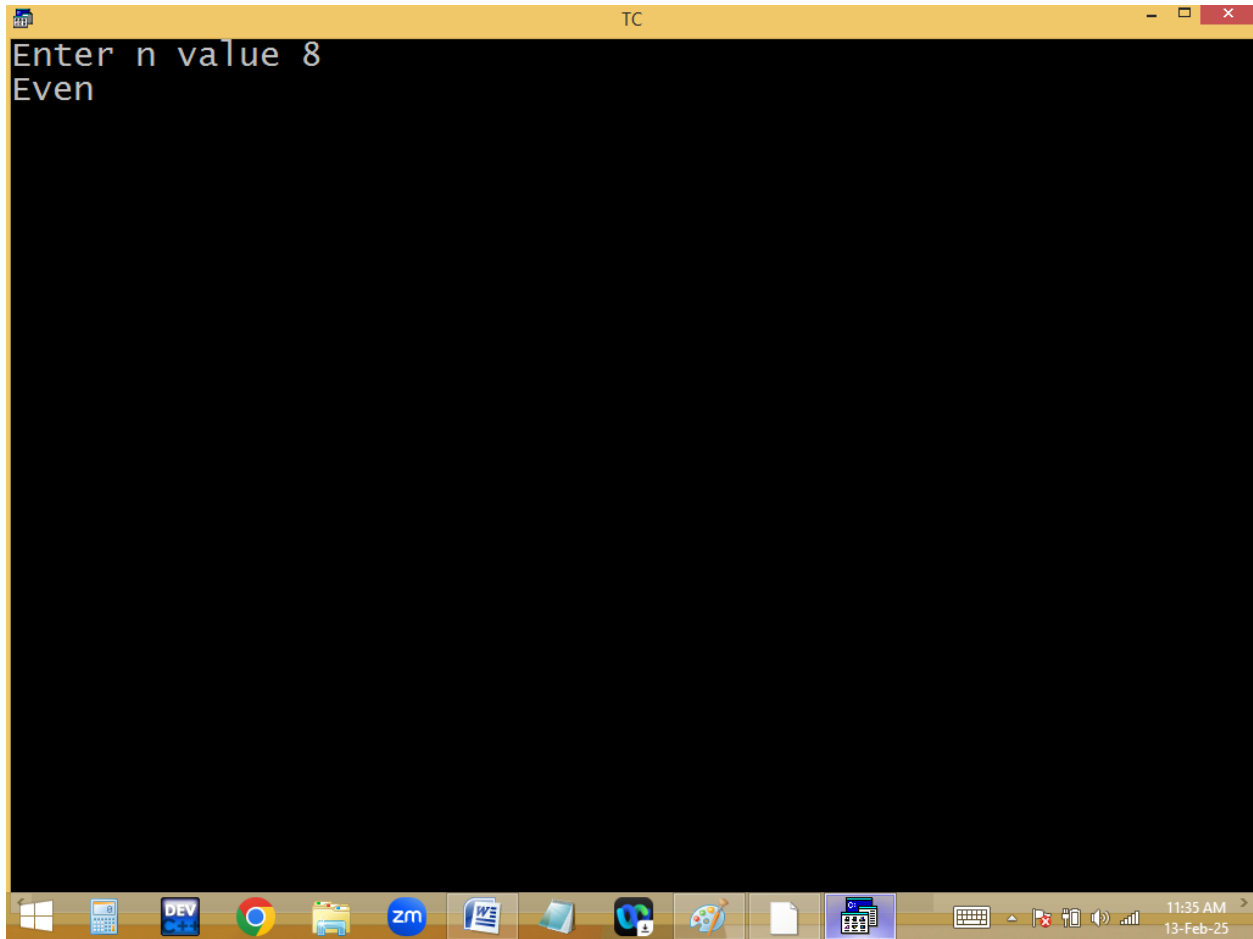


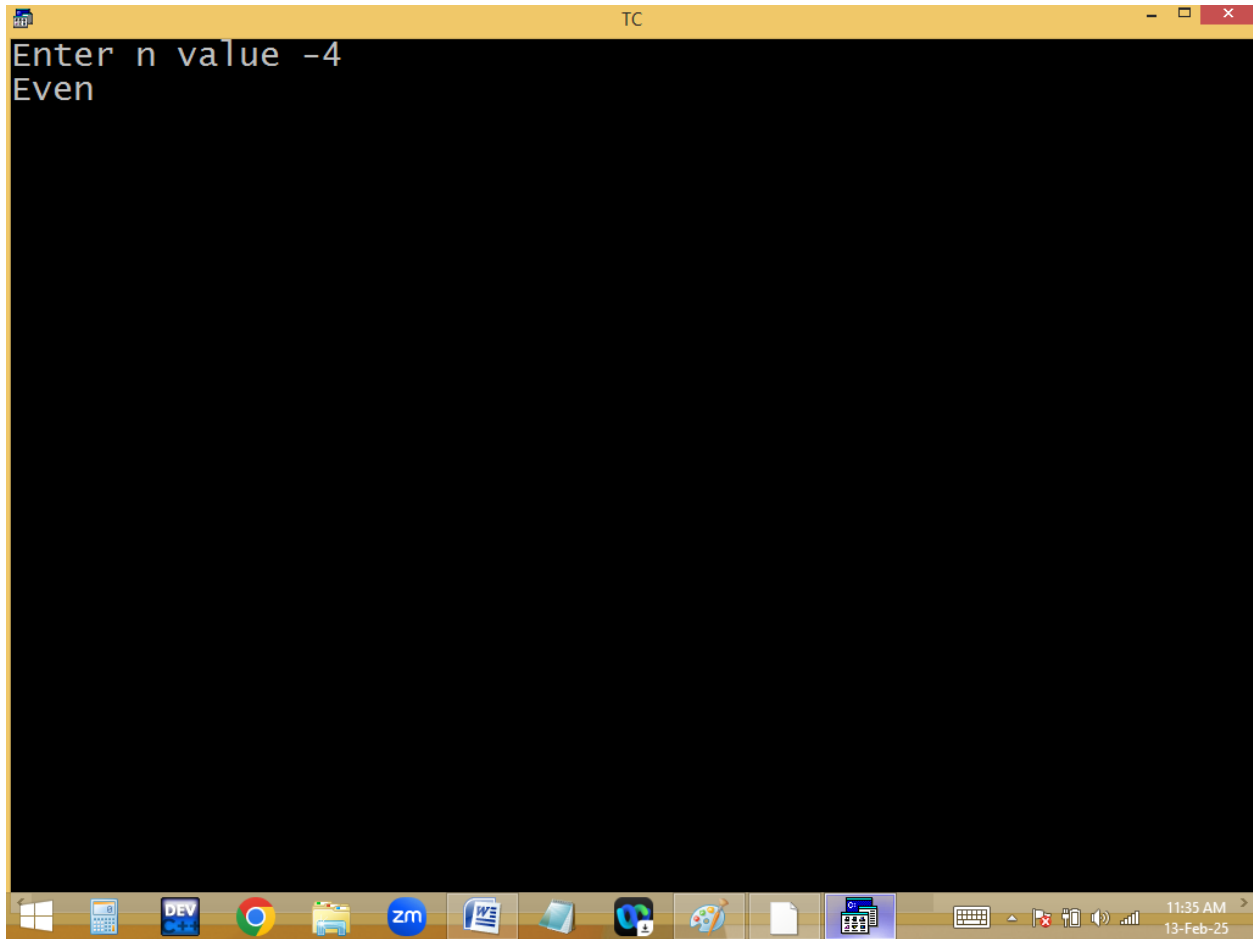
The image shows a screenshot of the Turbo C++ (TC) integrated development environment. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 9", "Col 27", and a list of editing commands: "Insert", "Indent", "Tab", "Fill", "Unindent", and a help key. The main editing area has a dark blue background with yellow text. The code is a C program that checks if a number is even or odd. The code is as follows:

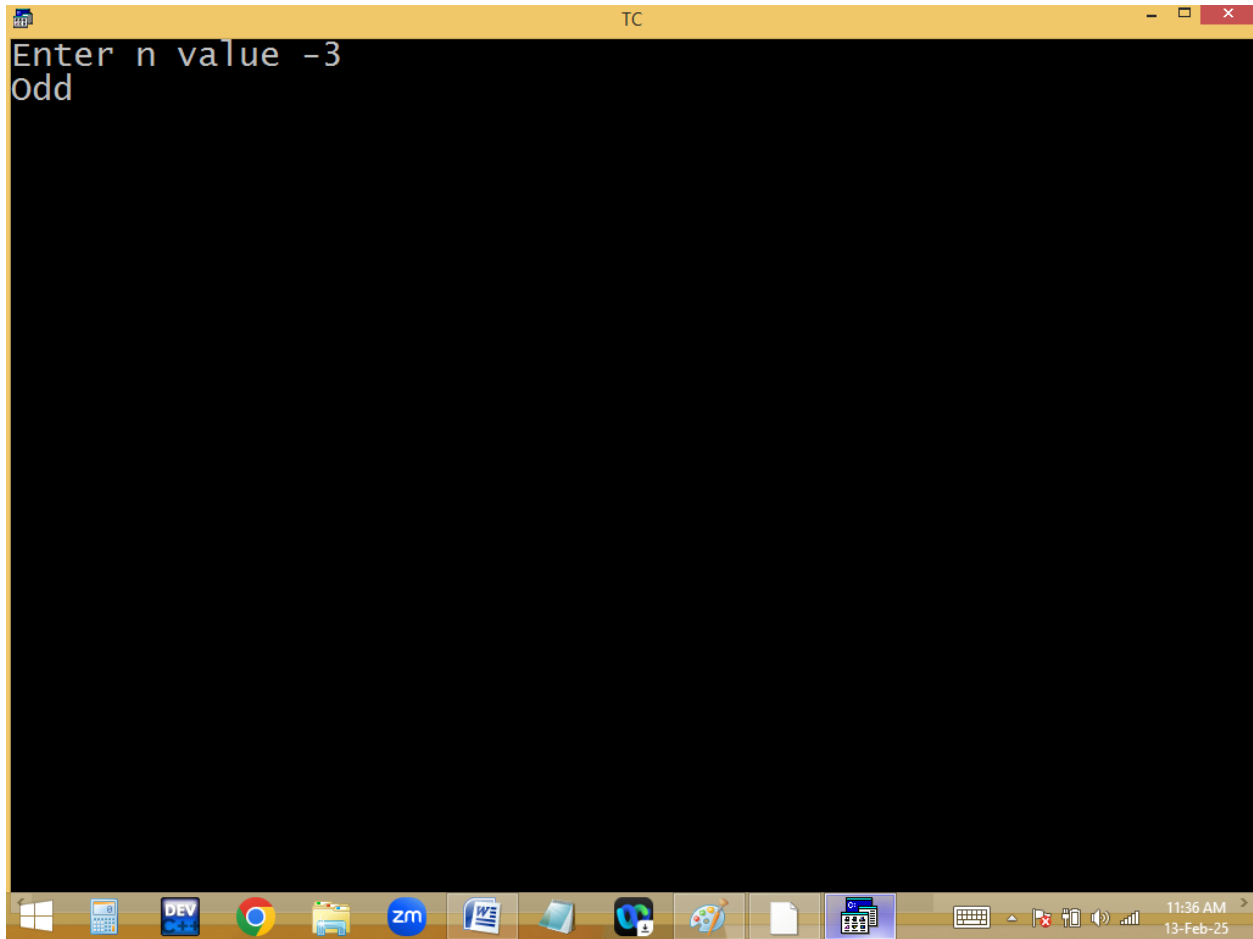
```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value ");
scanf("%d",&n);
n%2==0 && printf("Even")||_printf("Odd");
getch();
}
```

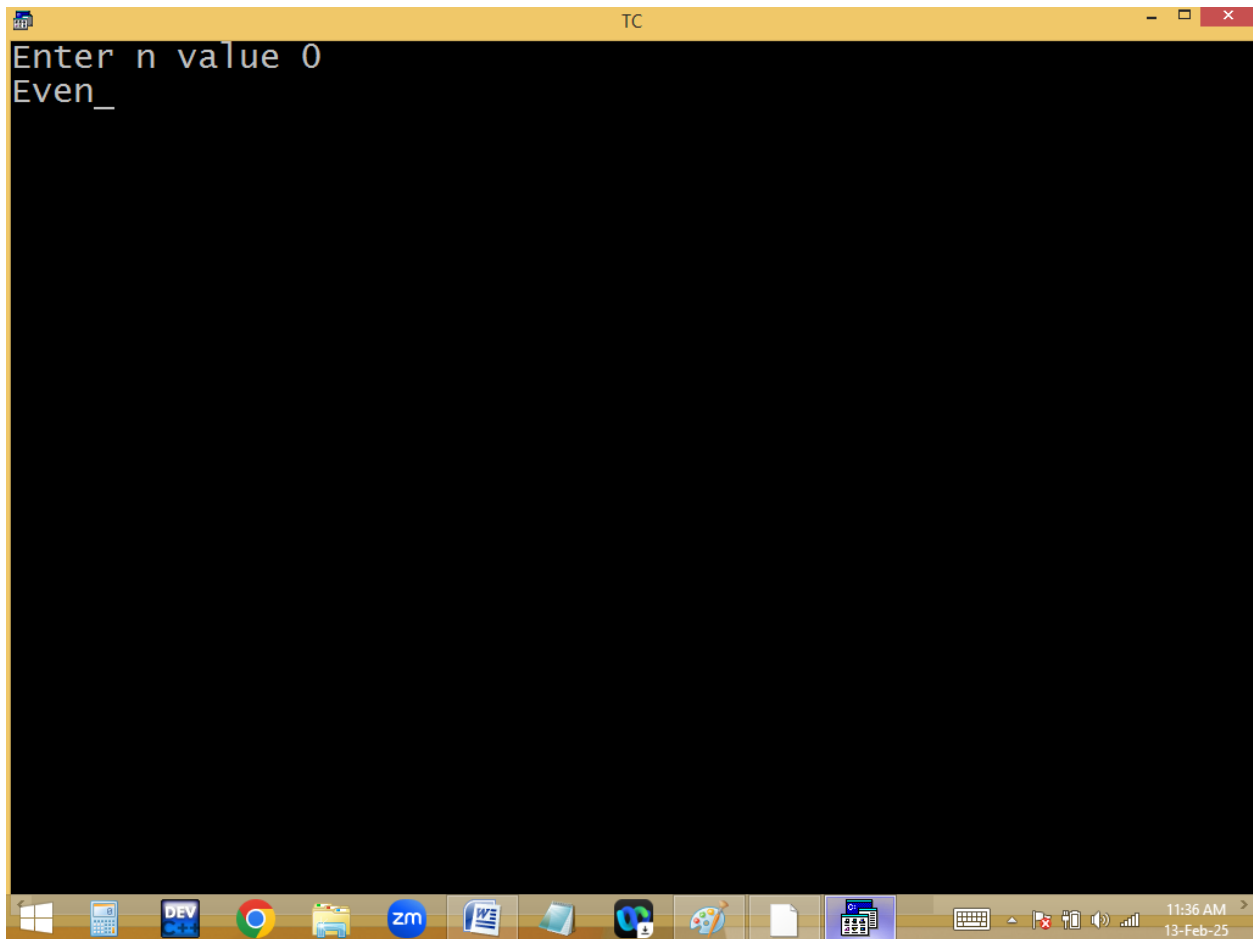
At the bottom of the window, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the very bottom, showing various application icons and the system clock indicating 11:35 AM on 13-Feb-25.











`if(n%2==0) p(even);`

`else p(odd);`

`4%2=0==0`

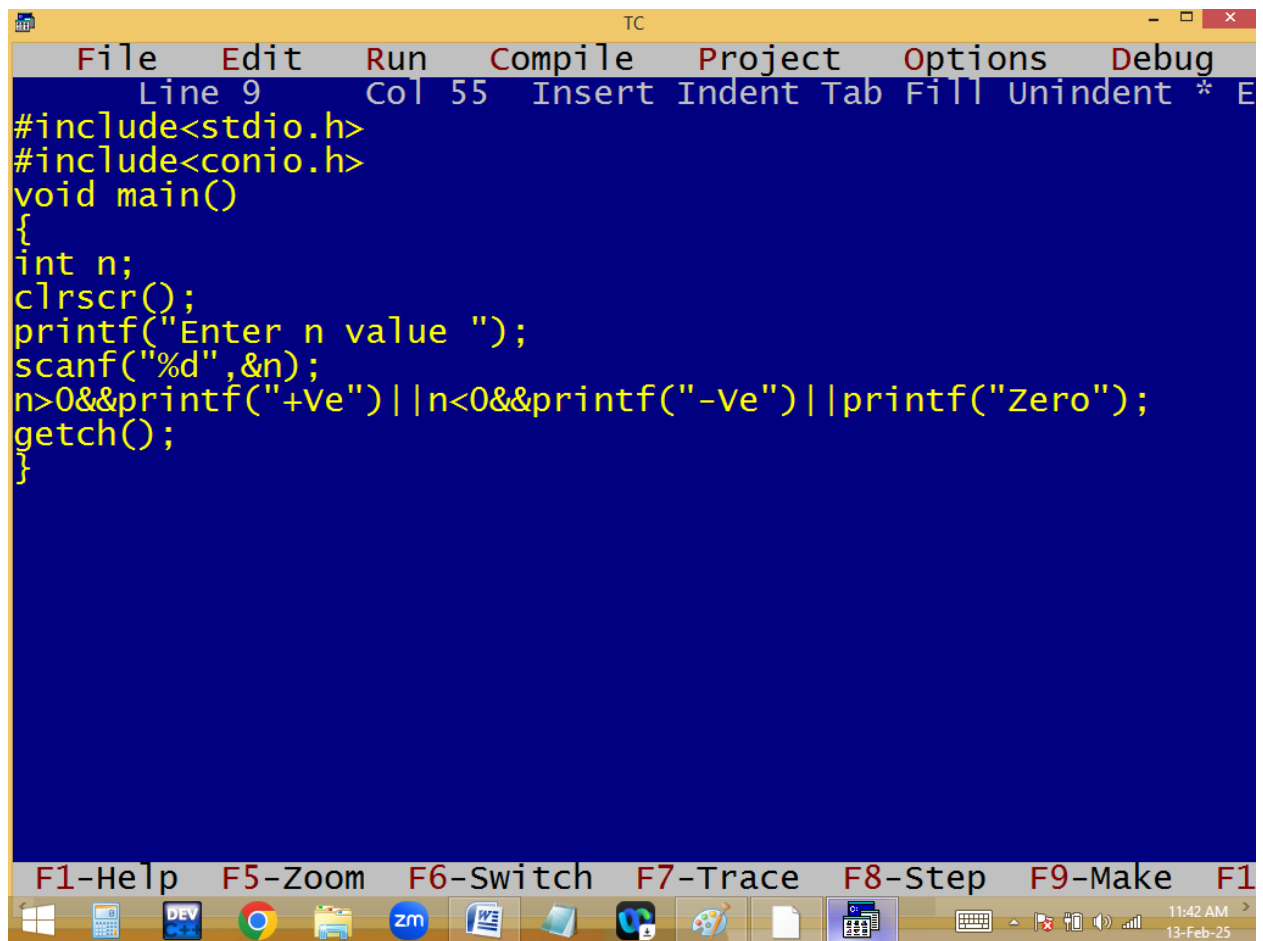


`n%2==0 && printf("Even") || printf("Odd");`

`5%2=1==0==>>false`



Finding +Ve/-Ve/0 without using if..else / ternary op:

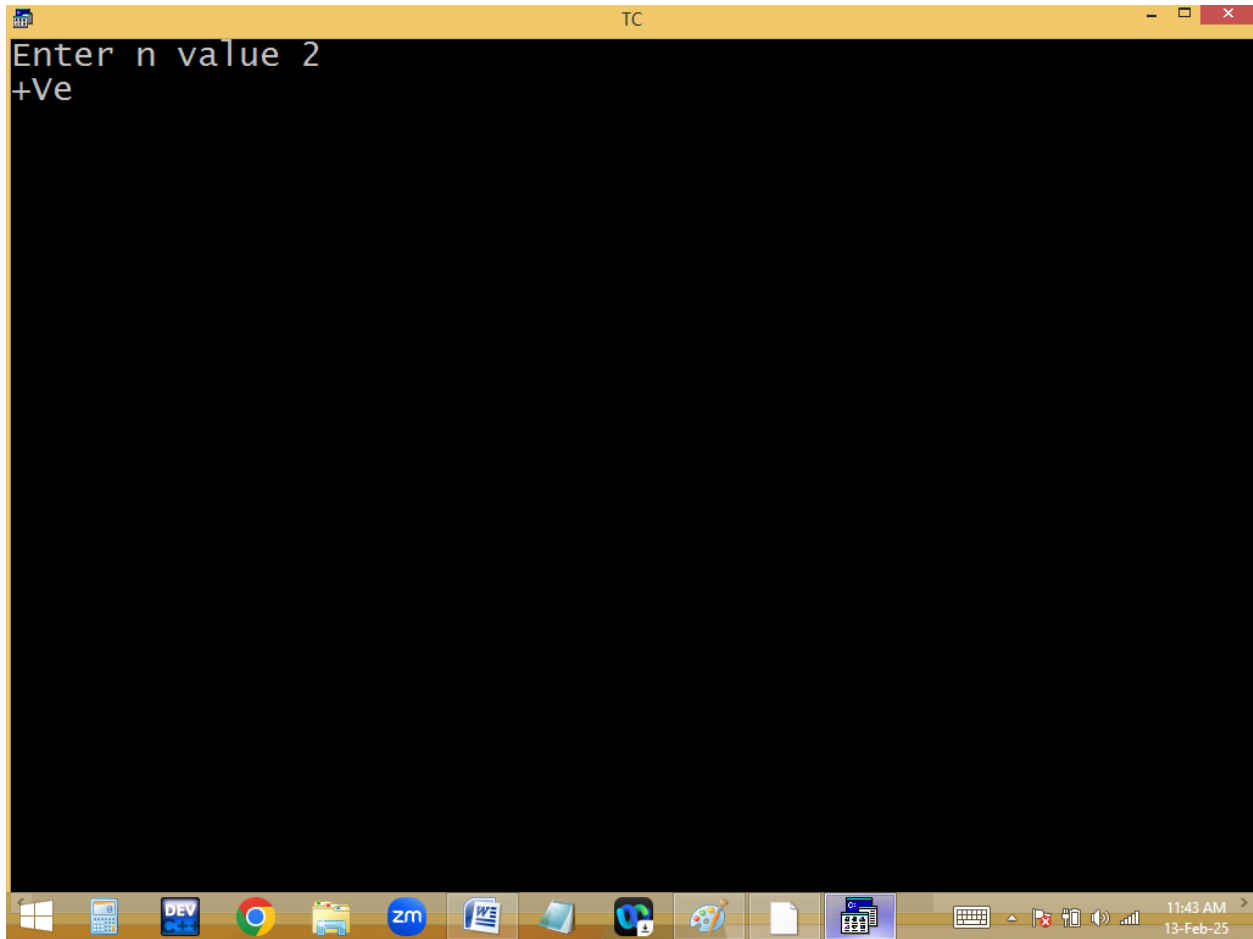


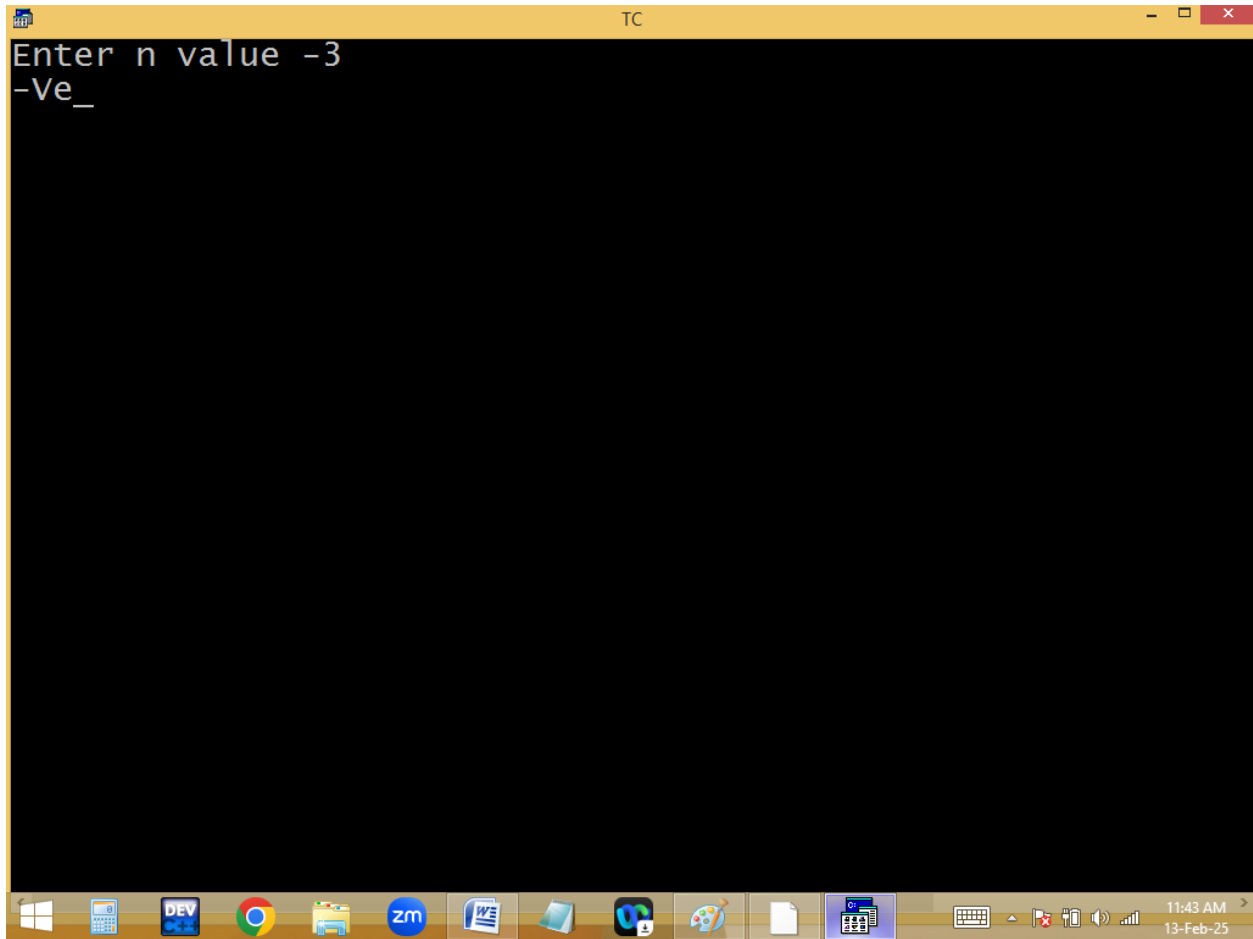
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following options: File, Edit, Run, Compile, Project, Options, and Debug. A status bar at the top of the editor area shows "Line 9", "Col 55", and a list of keyboard shortcuts: Insert, Indent, Tab, Fill, Unindent, *, and E. The main editor area has a dark blue background with yellow text. The code is as follows:

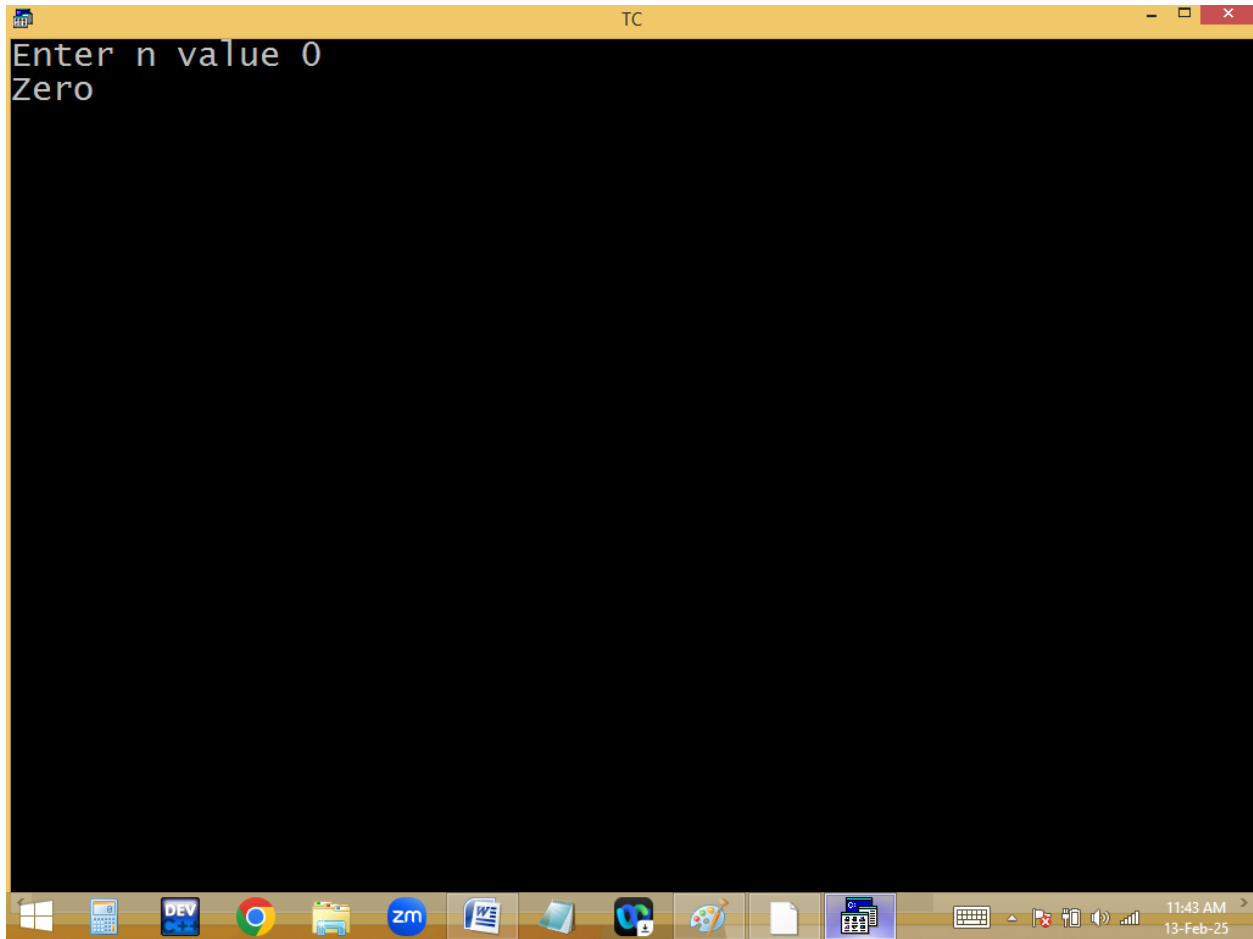
```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value ");
scanf("%d",&n);
n>0&&printf("+Ve")||n<0&&printf("-Ve")||printf("Zero");
getch();
}
```

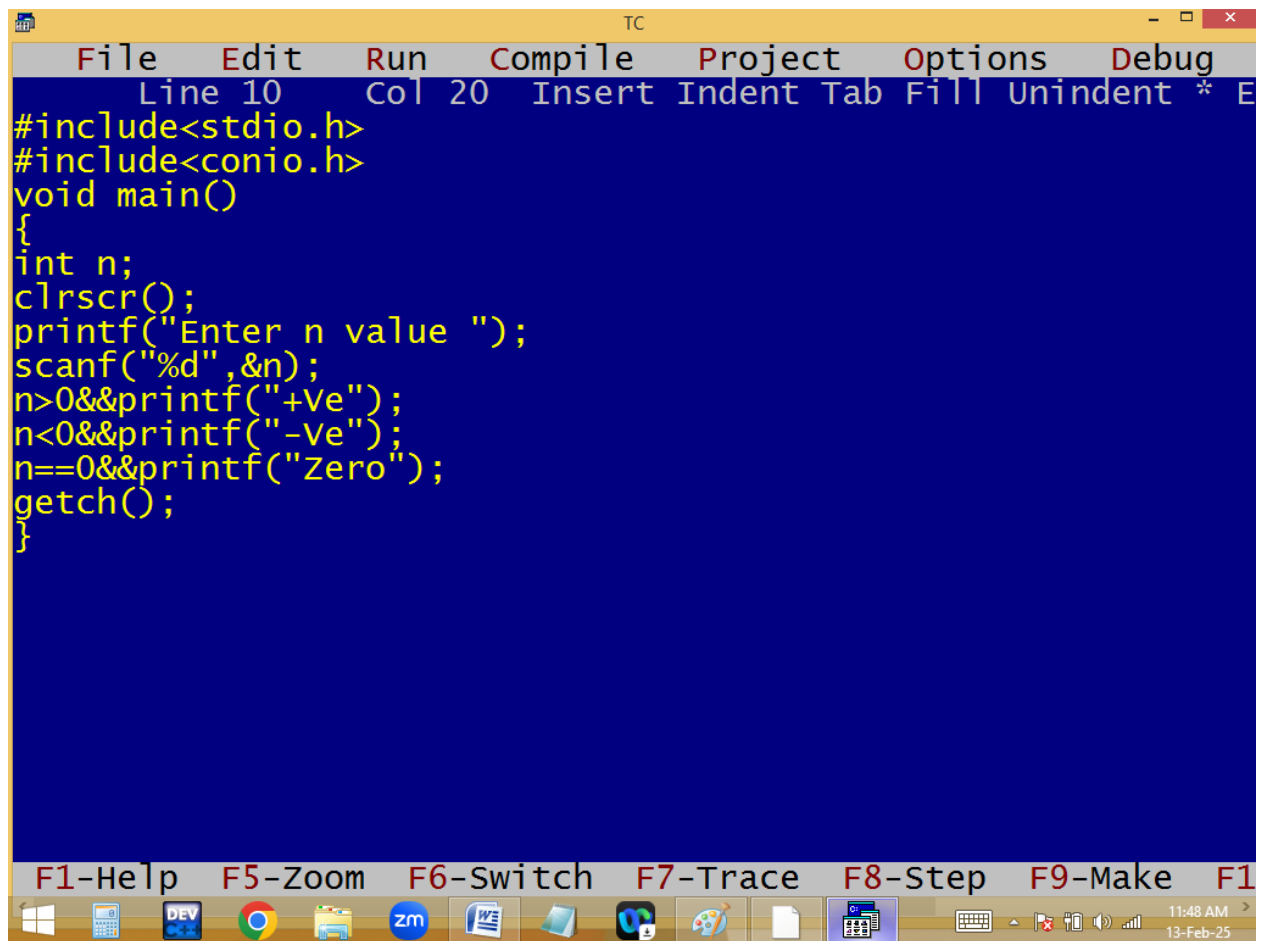
Below the editor area is a toolbar with various icons for file operations and editing. At the very bottom of the window is a Windows taskbar with several application icons, including the Start button, Task View, File Explorer, and others. The system clock in the bottom right corner shows "11:42 AM" and "13-Feb-25".

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run





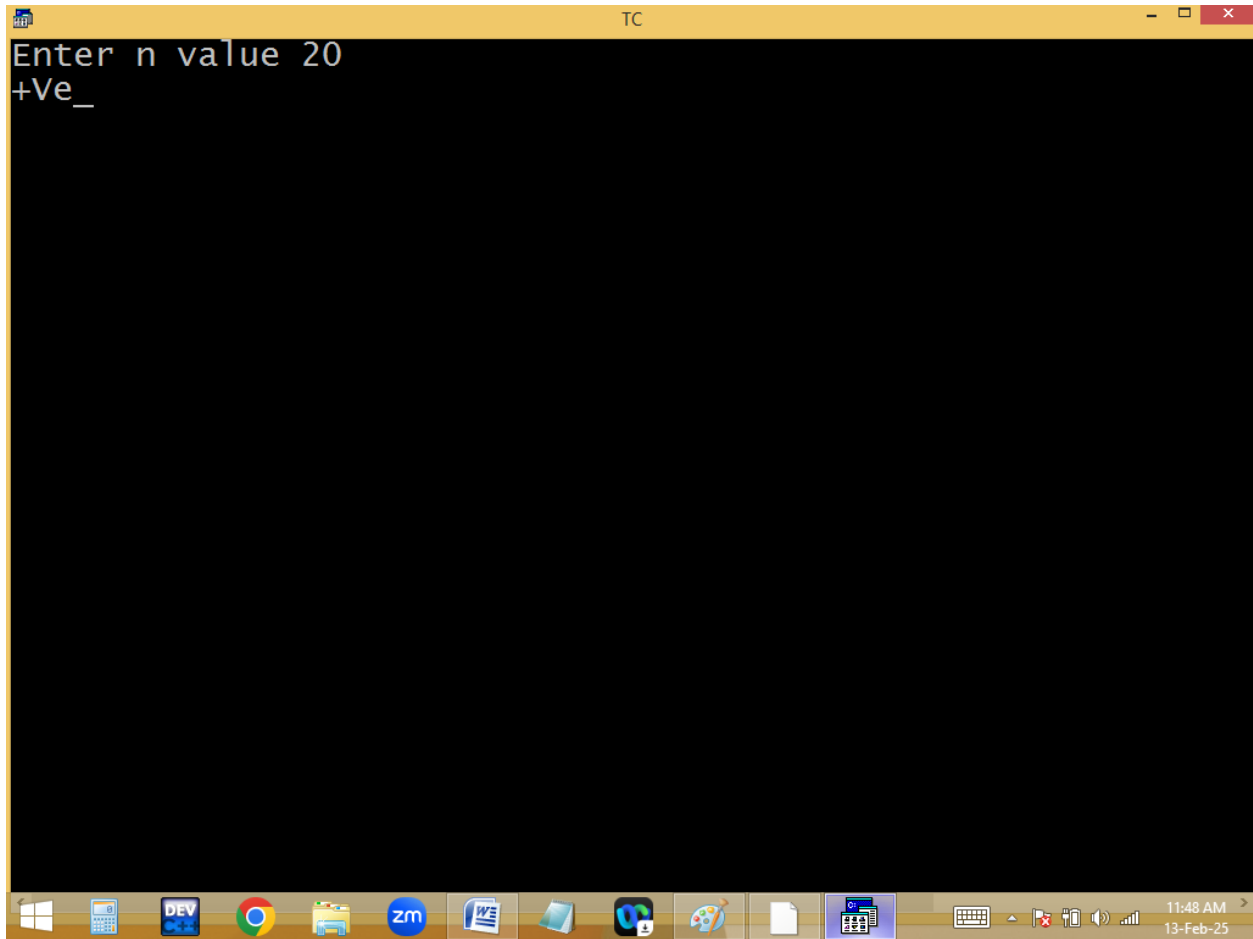


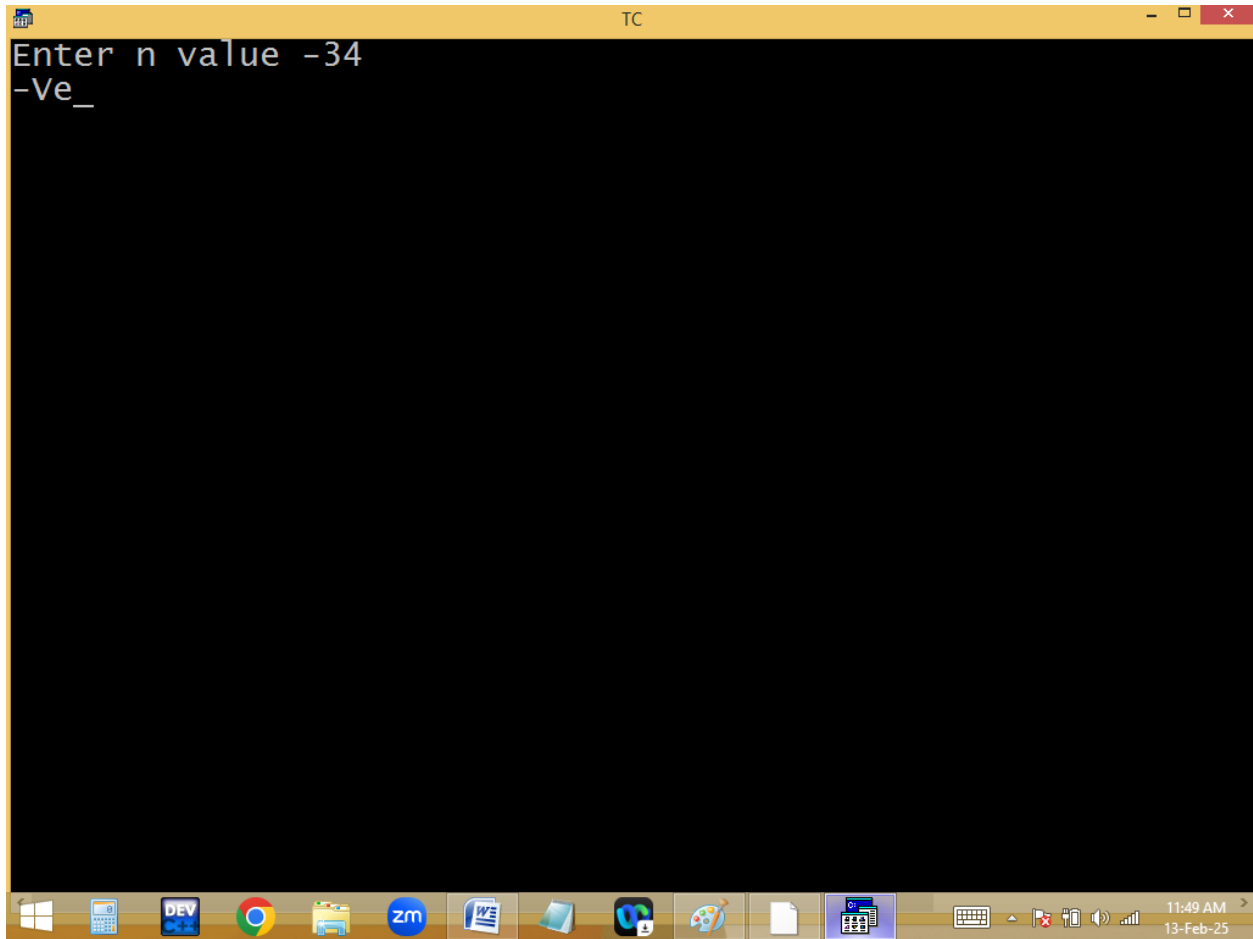


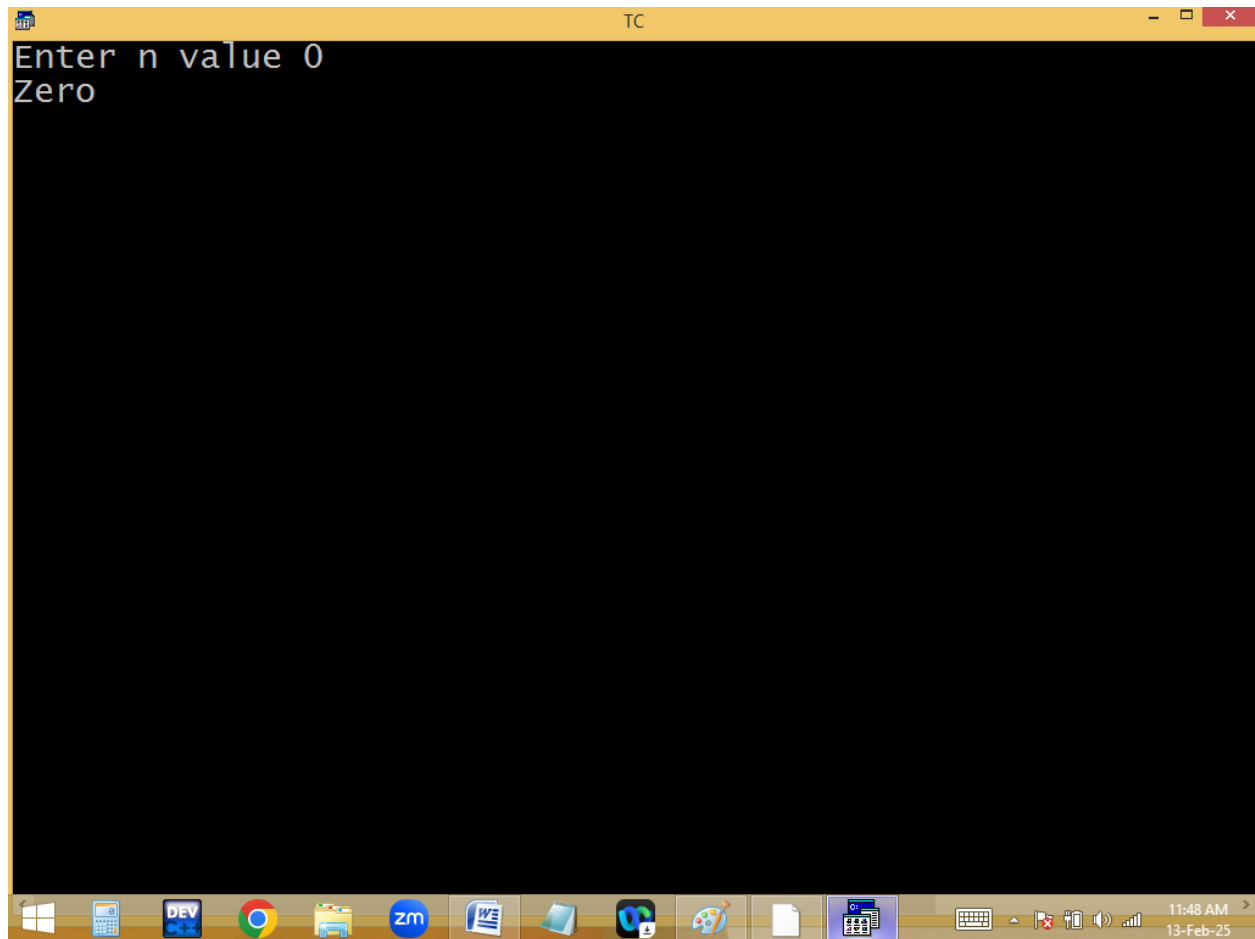
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 10", "Col 20", and various editing options like "Insert", "Indent", "Tab", "Fill", "Unindent", and a "E" (likely for End). The main editing area has a dark blue background with yellow text. The code is a C program that prompts the user to enter a value 'n' and then checks if it is greater than zero, less than zero, or equal to zero, printing the corresponding result. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value ");
scanf("%d",&n);
n>0&&printf("+ve");
n<0&&printf("-ve");
n==0&&printf("Zero");
getch();
}
```

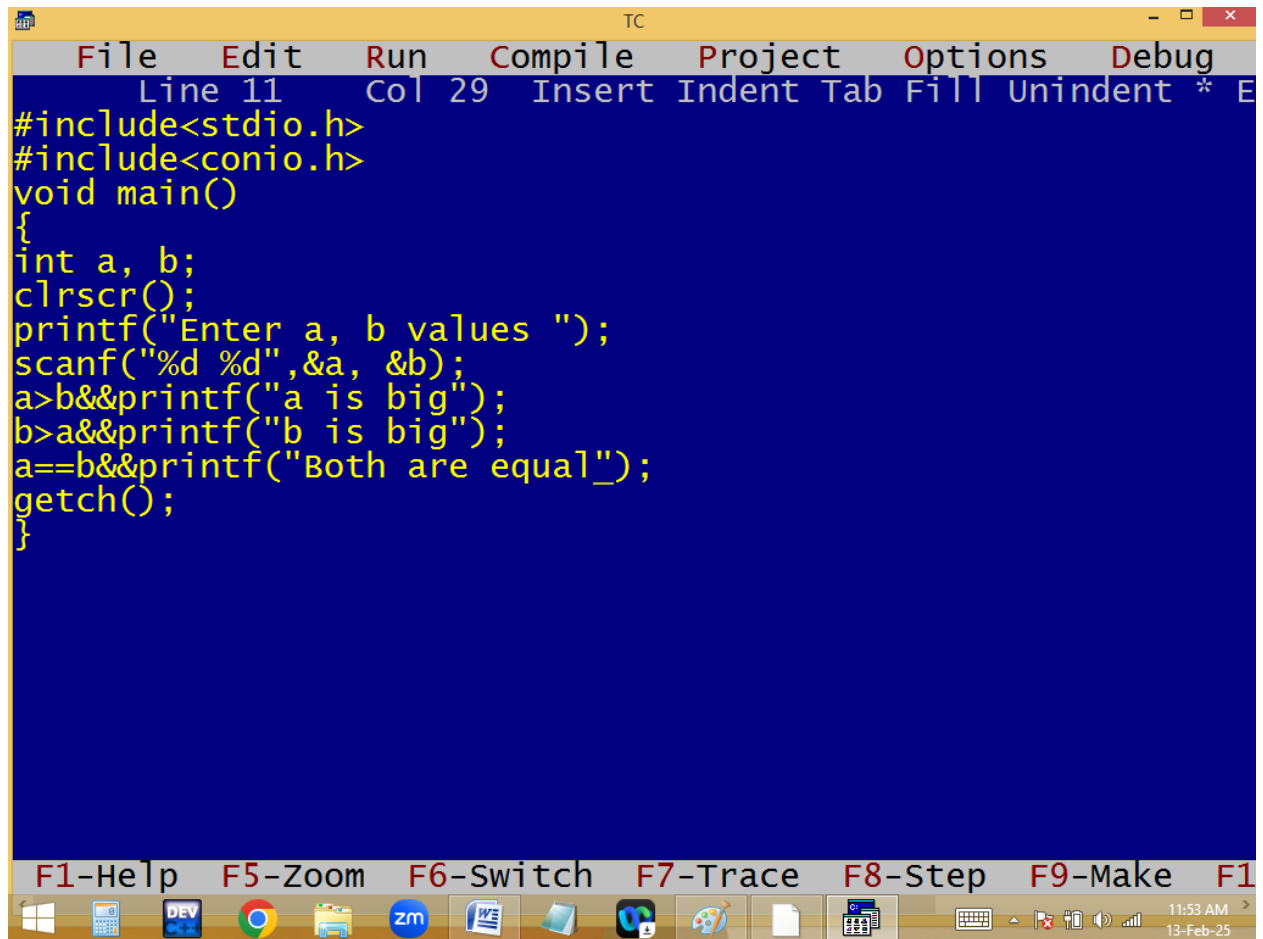
At the bottom of the window, there is a toolbar with icons for various functions, each with a corresponding function key label: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the very bottom, showing several open applications and the system clock indicating 11:48 AM on 13-Feb-25.







Finding max in 2 no's without using if..else / ternary op:

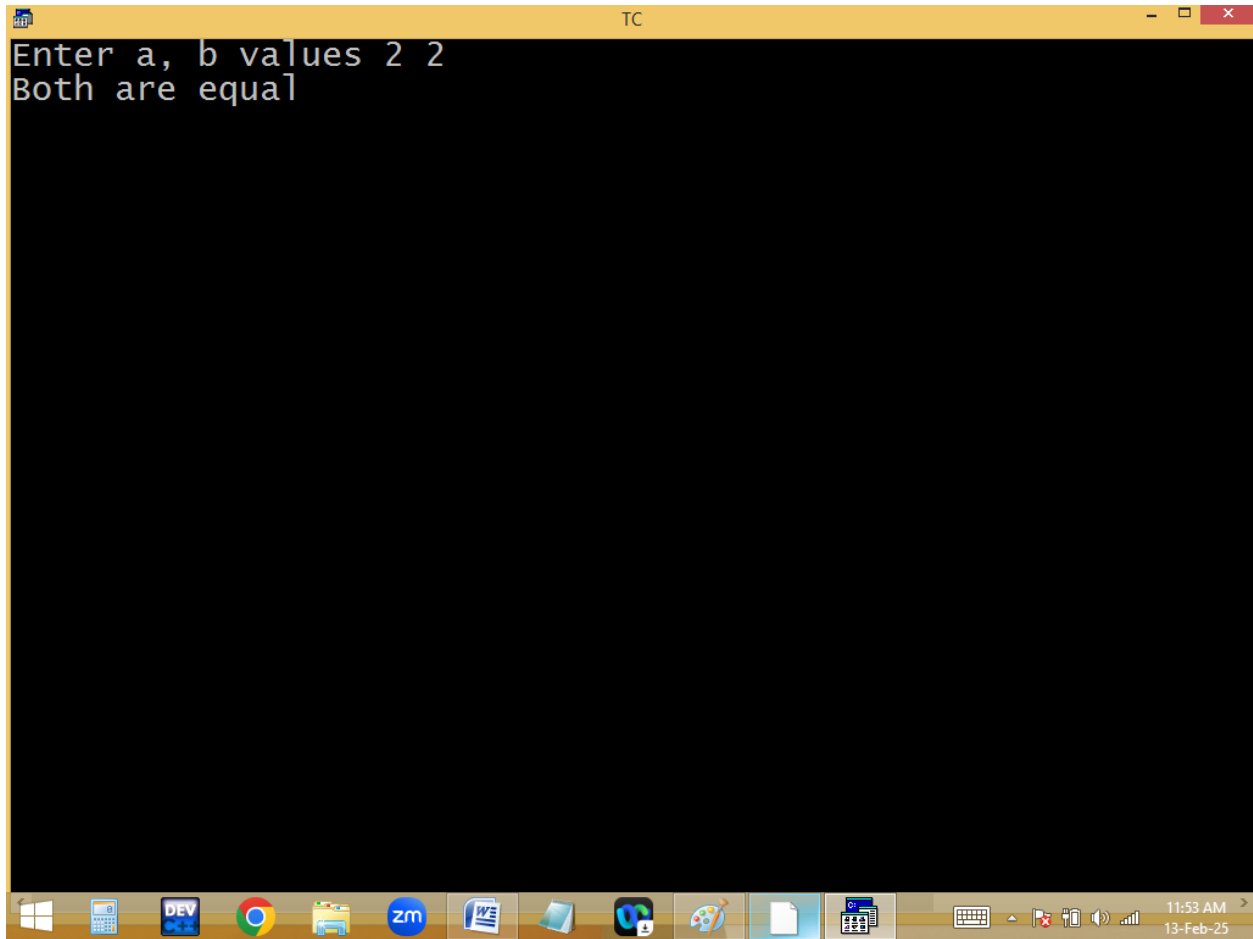


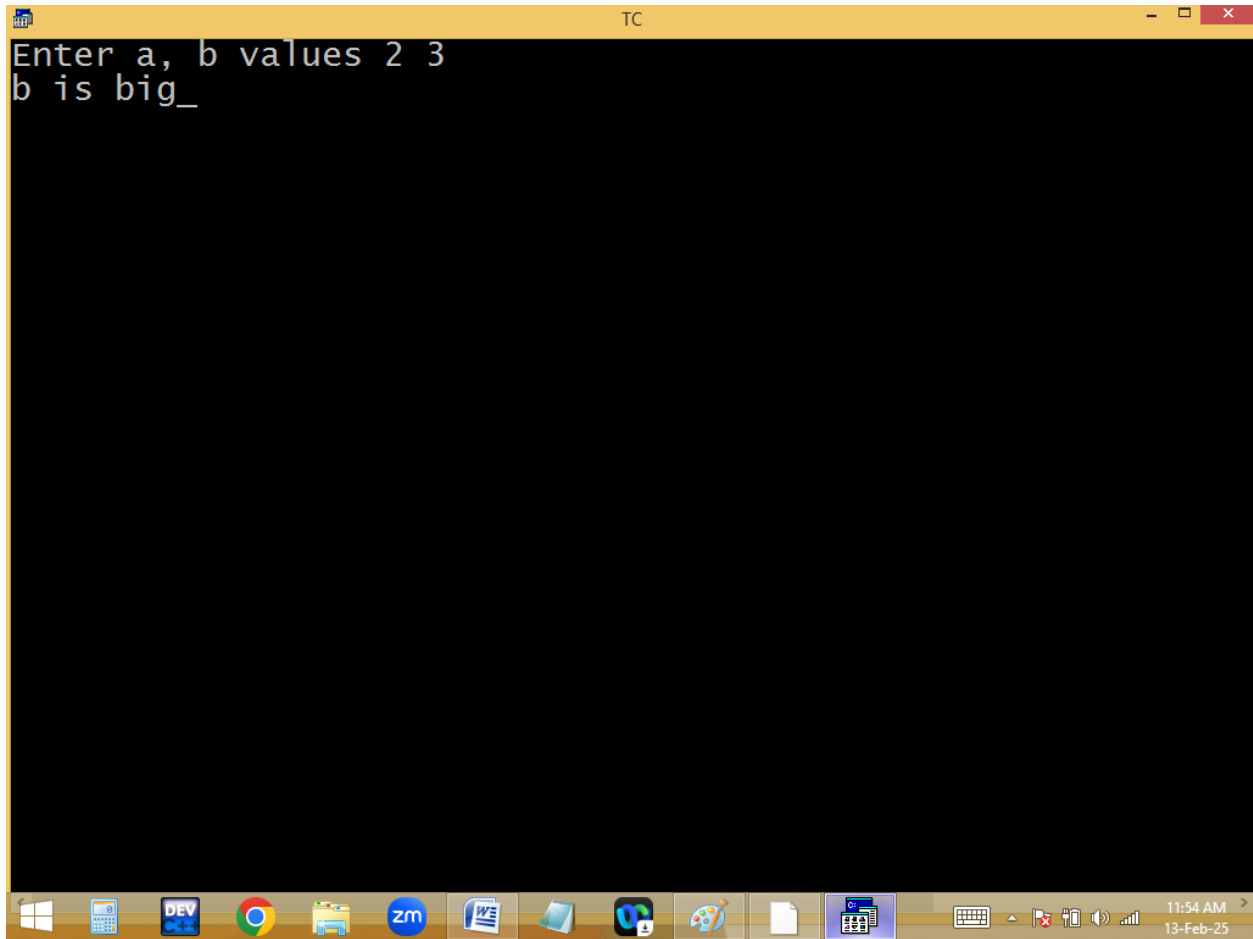
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 11", "Col 29", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and "* E". The main editing area has a dark blue background with yellow text. The code is a C program that compares two integers, a and b. It includes headers for stdio.h and conio.h, defines a main function, declares variables a and b, clears the screen, prompts the user for input, reads the input using scanf, and then uses printf to output the comparison results. The program ends with getch() and a closing brace for main. At the bottom of the window, there is a toolbar with icons for various functions and a row of function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F10-Run. The Windows taskbar is visible at the very bottom, showing the Start button, taskbar icons for Windows Explorer, Dev-C++, Google Chrome, File Explorer, Zoom, Word, and others, along with the system clock showing 11:53 AM on 13-Feb-25.

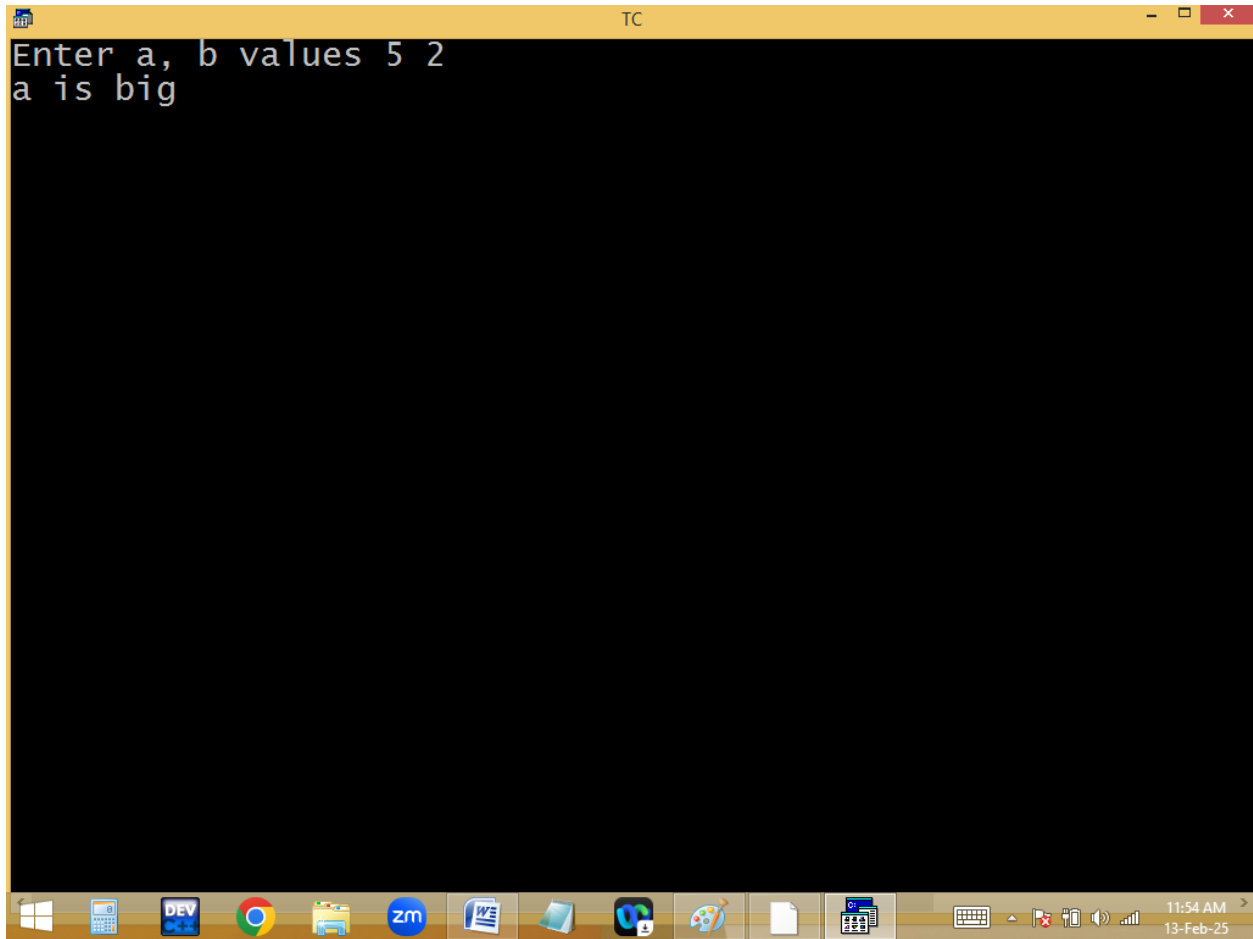
```
File Edit Run Compile Project Options Debug
Line 11 Col 29 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b;
clrscr();
printf("Enter a, b values ");
scanf("%d %d",&a, &b);
a>b&&printf("a is big");
b>a&&printf("b is big");
a==b&&printf("Both are equal");
getch();
}
```

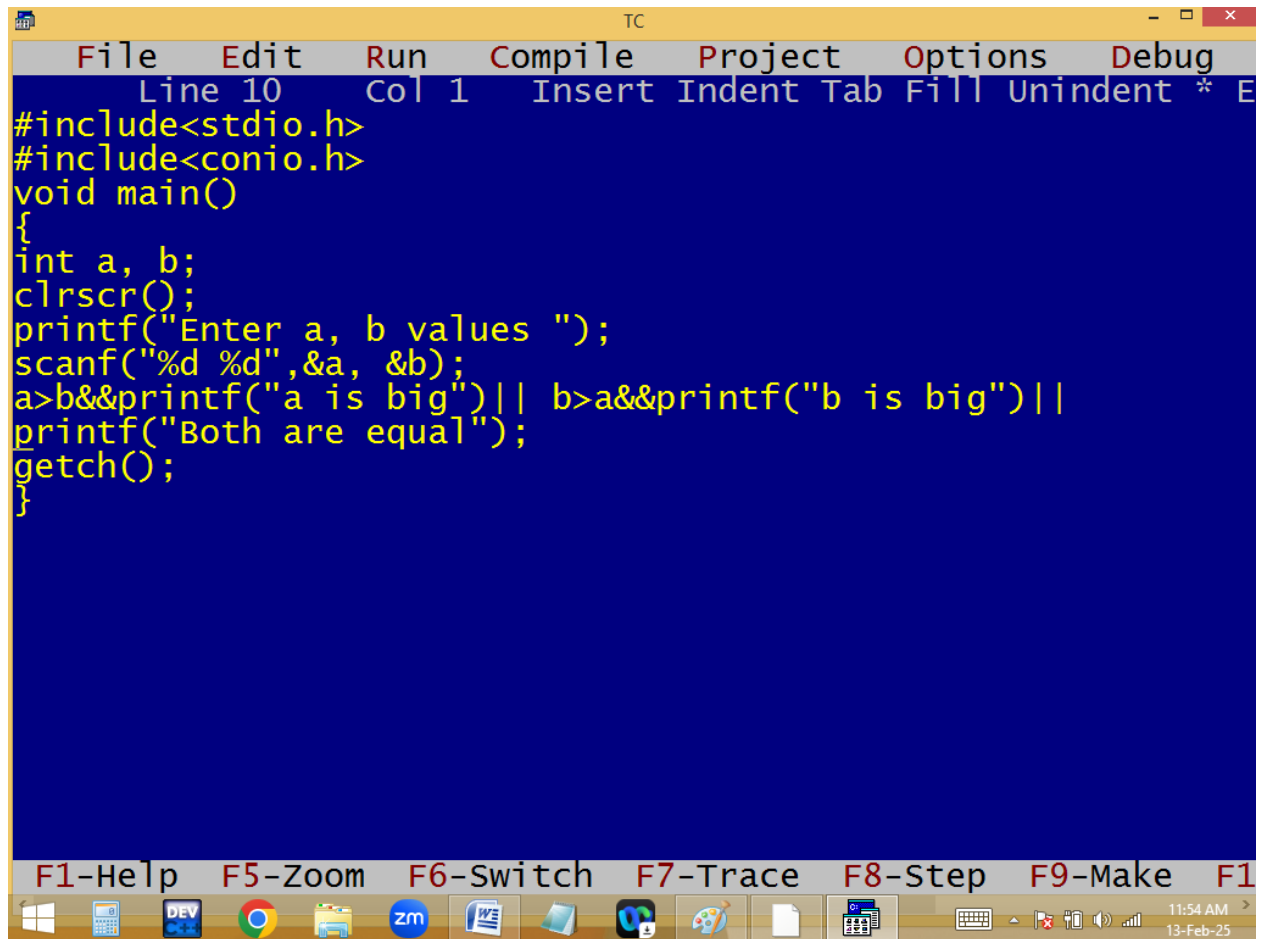
F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Run

11:53 AM
13-Feb-25





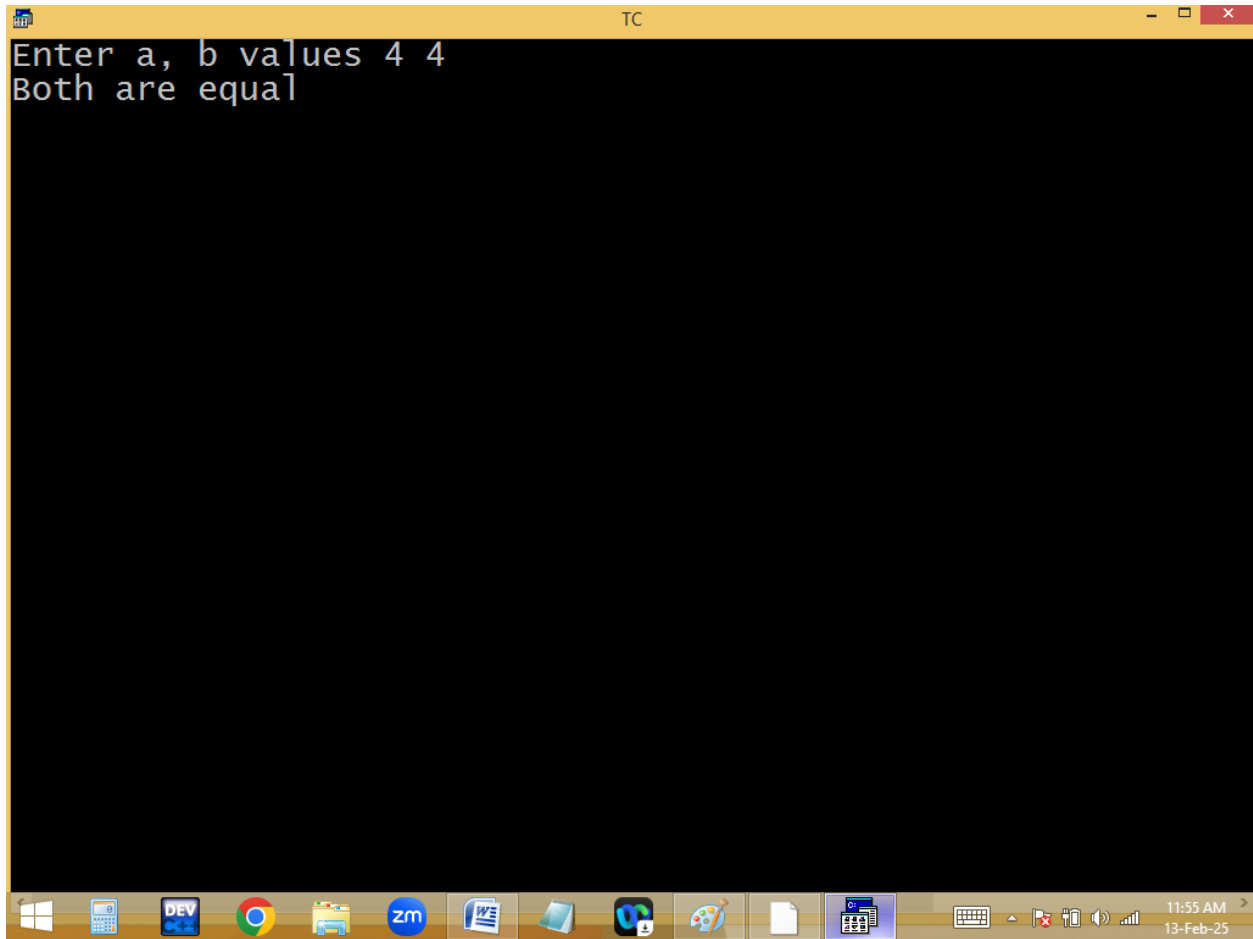


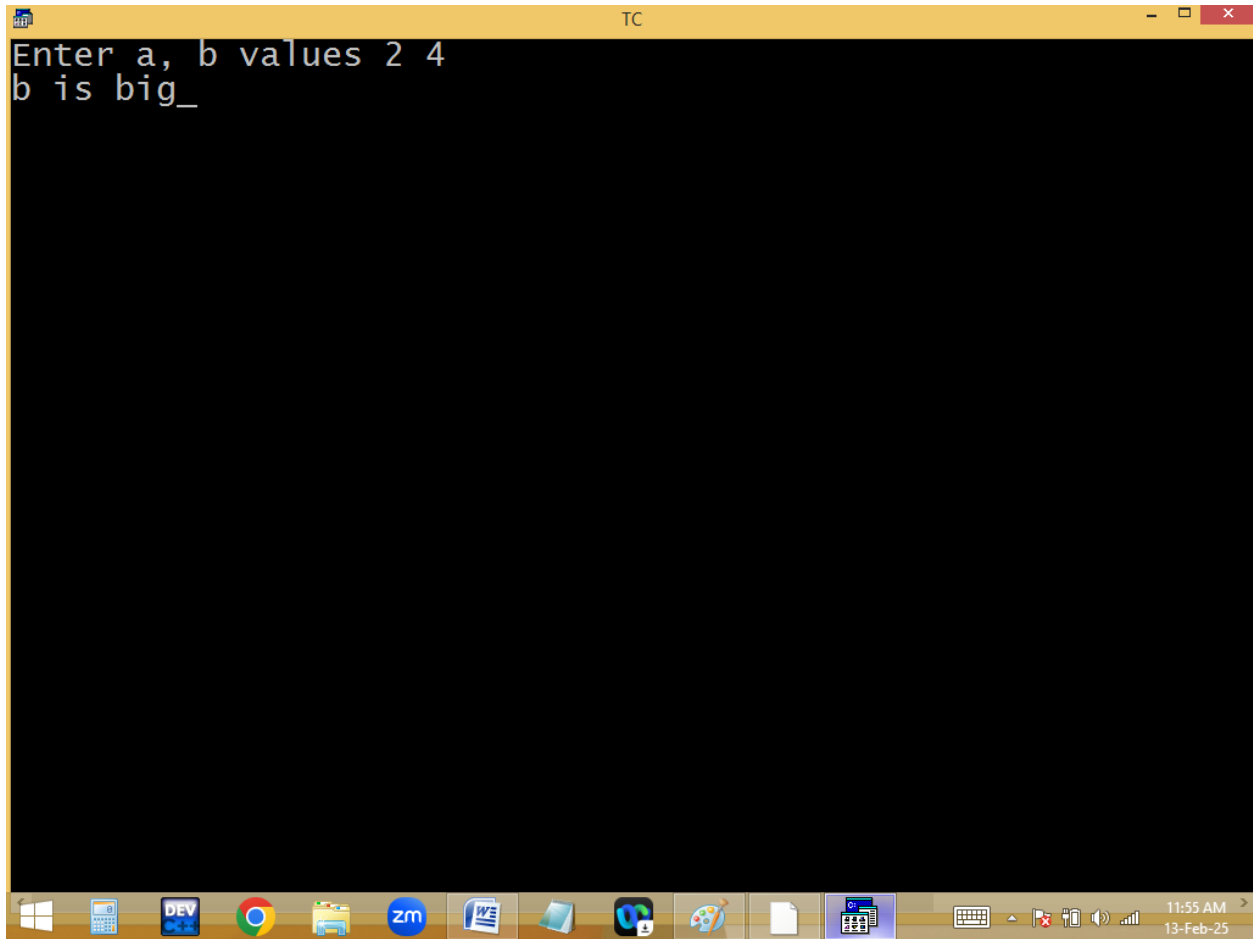


The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 10", "Col 1", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and "* E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int a, b;
clrscr();
printf("Enter a, b values ");
scanf("%d %d",&a, &b);
a>b&&printf("a is big")|| b>a&&printf("b is big")||
printf("Both are equal");
getch();
}
```

At the bottom of the window, there is a toolbar with icons for various functions. Below the toolbar, a row of function key shortcuts is displayed: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Exit". The Windows taskbar is visible at the very bottom, showing icons for the Start menu, Task View, File Explorer, Google Chrome, and other applications. The system clock in the bottom right corner indicates the time is 11:54 AM on 13-Feb-25.





```
TC
Enter a, b values 20 1
a is big
```

3 > 8
5 > 2
if(a>b)p(a big); else if(b>a) p(b big); else p(equal);
3>3 =====> 3>3 =====> →

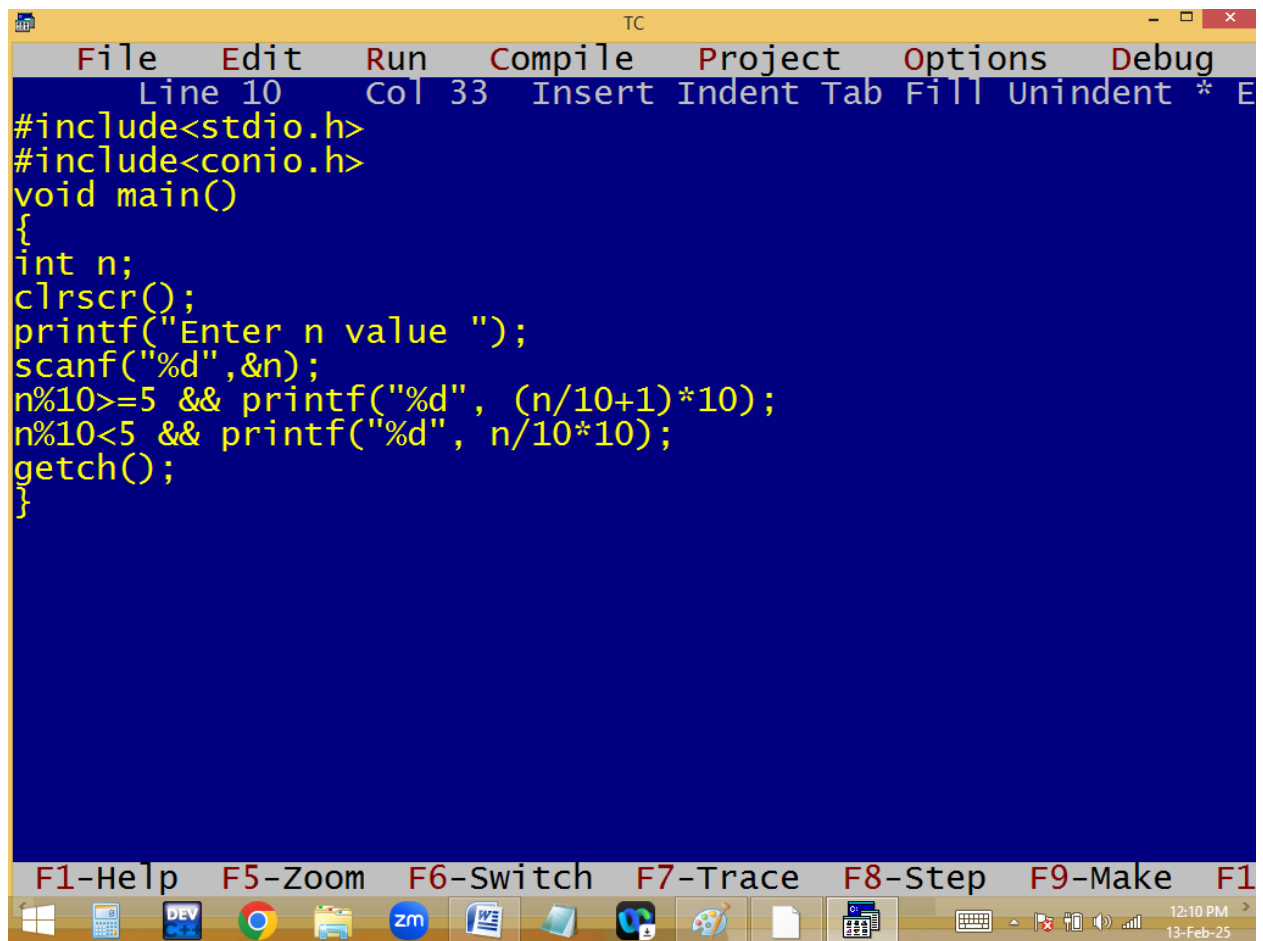
a>b&& p(a is big) || b>a&& p("b is big") || p(equal);
8>8 false || 8>8 false || equal
9>3 true a is big
4 > 7 false || 7>4 true b is big

Eg: rounding of no to its previous or next 10 multiple based on last digit without using if..else or ternary op.

24 last digit is 4, which is below 5 → 2 → 20

27 last digit is 7, which is above= 5 → $2+1=3*10=30$

25 last digit is 5, which is above= 5 → $2+1=3*10=30$

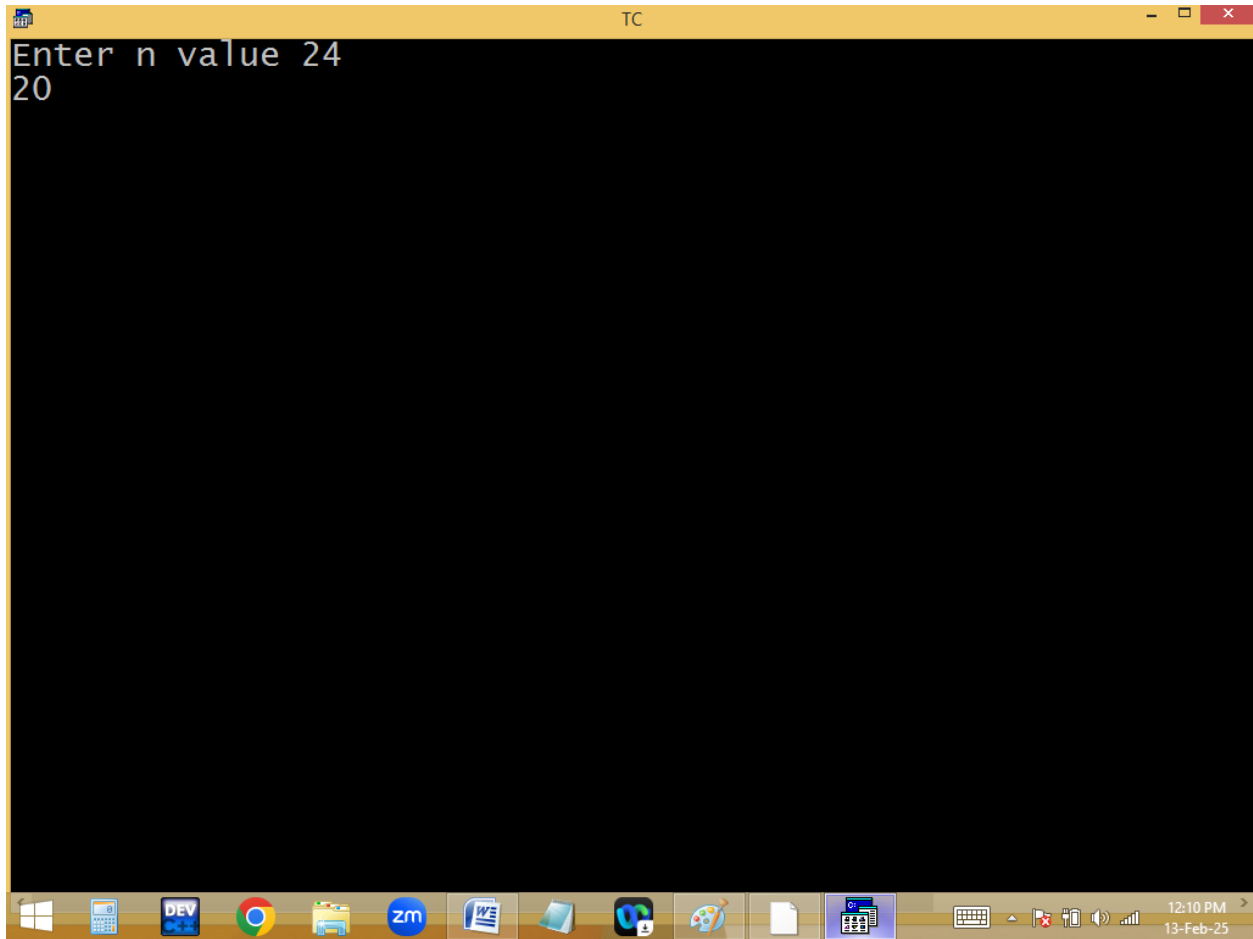


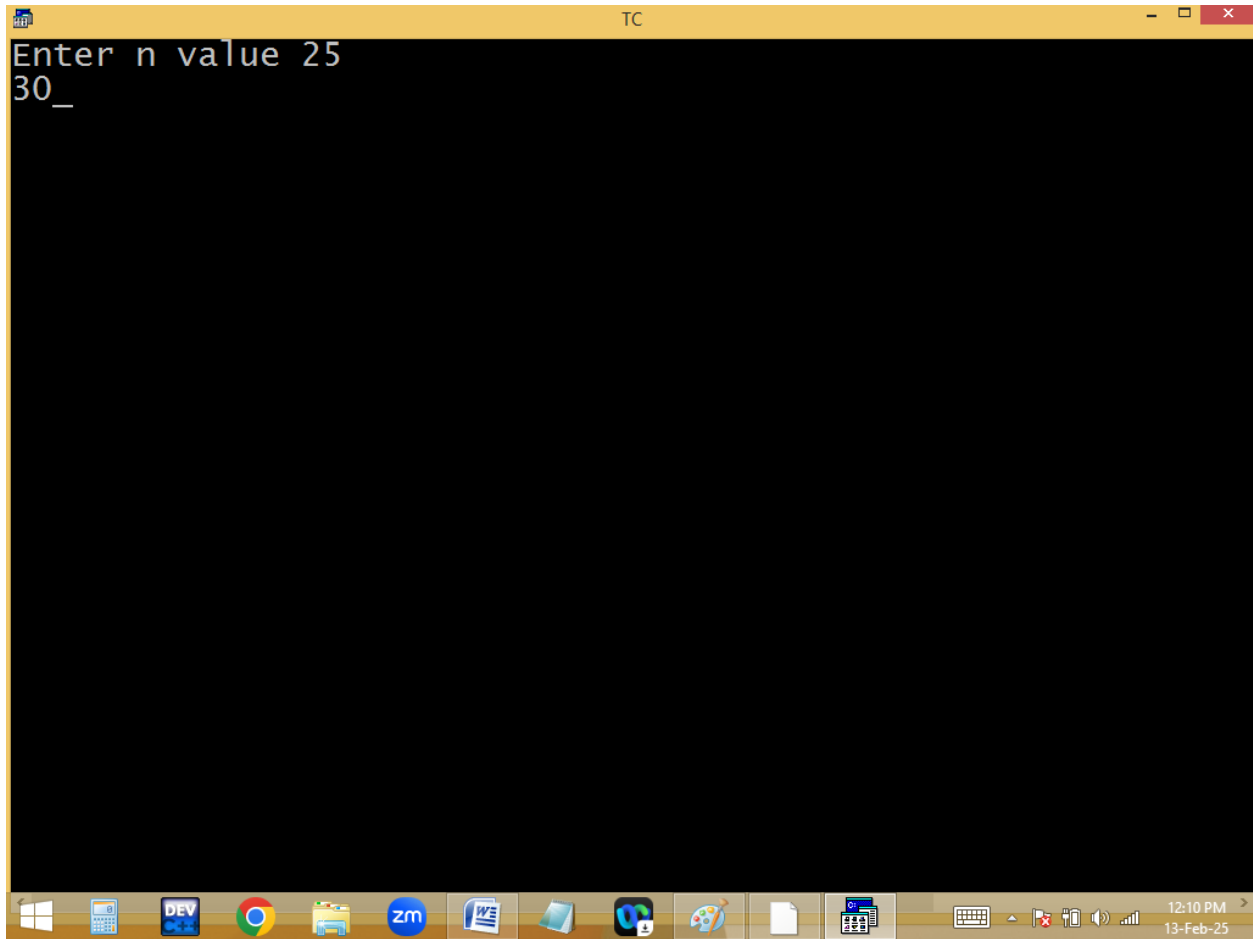
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. Below the title bar is a menu bar with the following items: File, Edit, Run, Compile, Project, Options, and Debug. Under the "Run" menu, there is a submenu with the following items: Line 10, Col 33, Insert, Indent, Tab, Fill, Unindent, and *. The main editing area has a dark blue background with yellow text. The code is as follows:

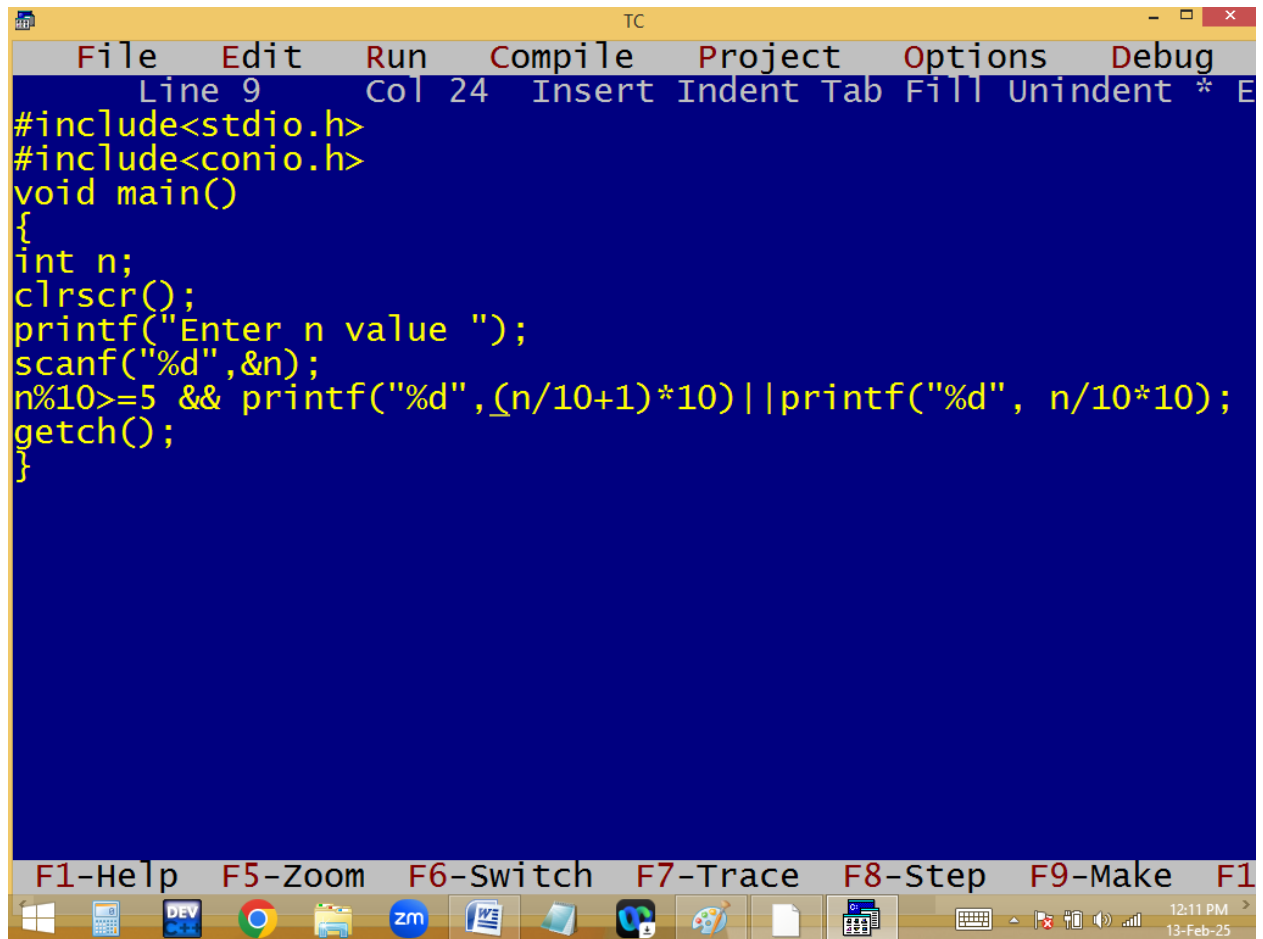
```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value ");
scanf("%d",&n);
n%10>=5 && printf("%d", (n/10+1)*10);
n%10<5 && printf("%d", n/10*10);
getch();
}
```

Below the editing area is a status bar with the following items: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, and F1. At the bottom of the window is a Windows taskbar with various icons, including the Start button, taskbar buttons for DEV, Google Chrome, File Explorer, Zoom, Word, and others. The system clock in the bottom right corner shows 12:10 PM on 13-Feb-25.





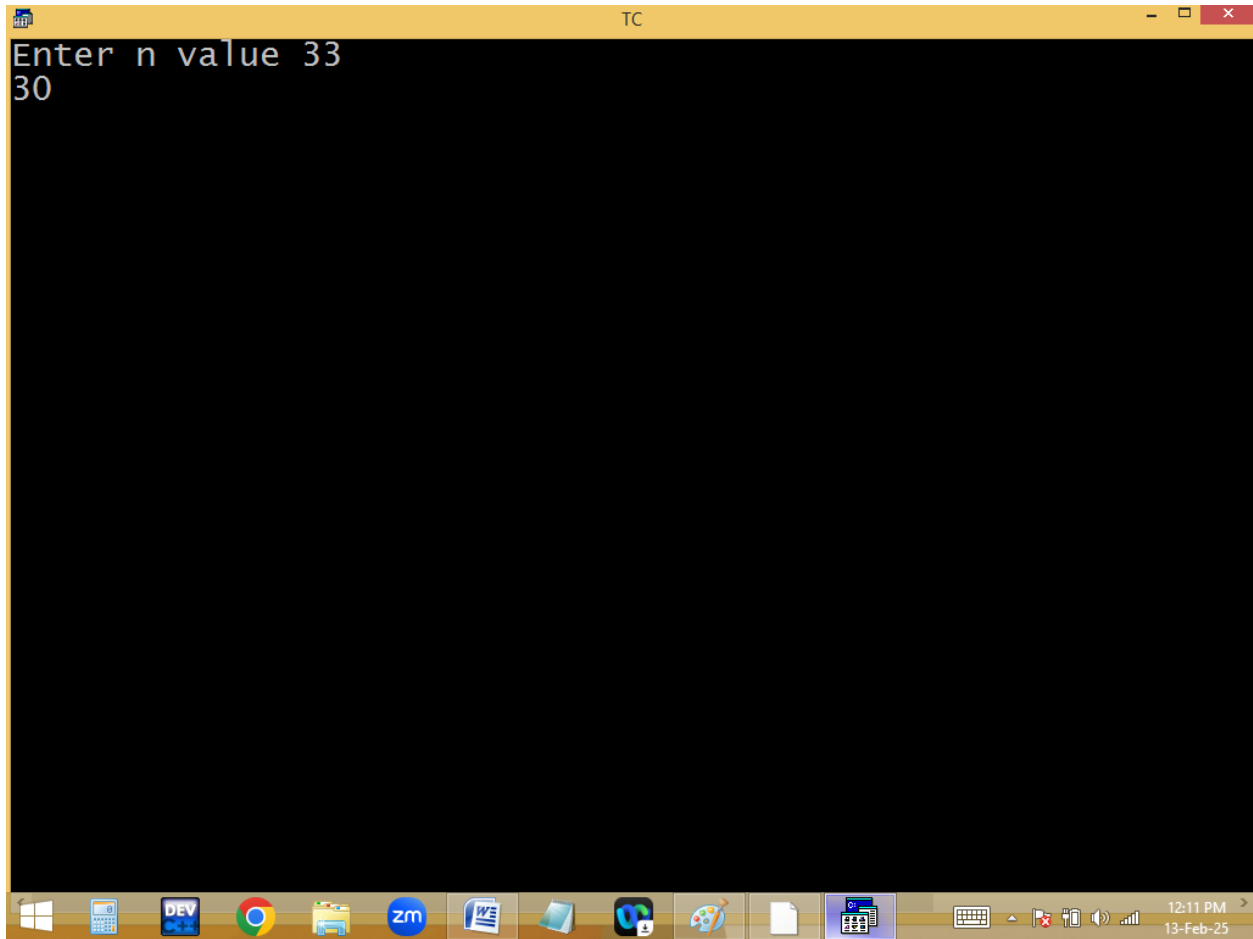


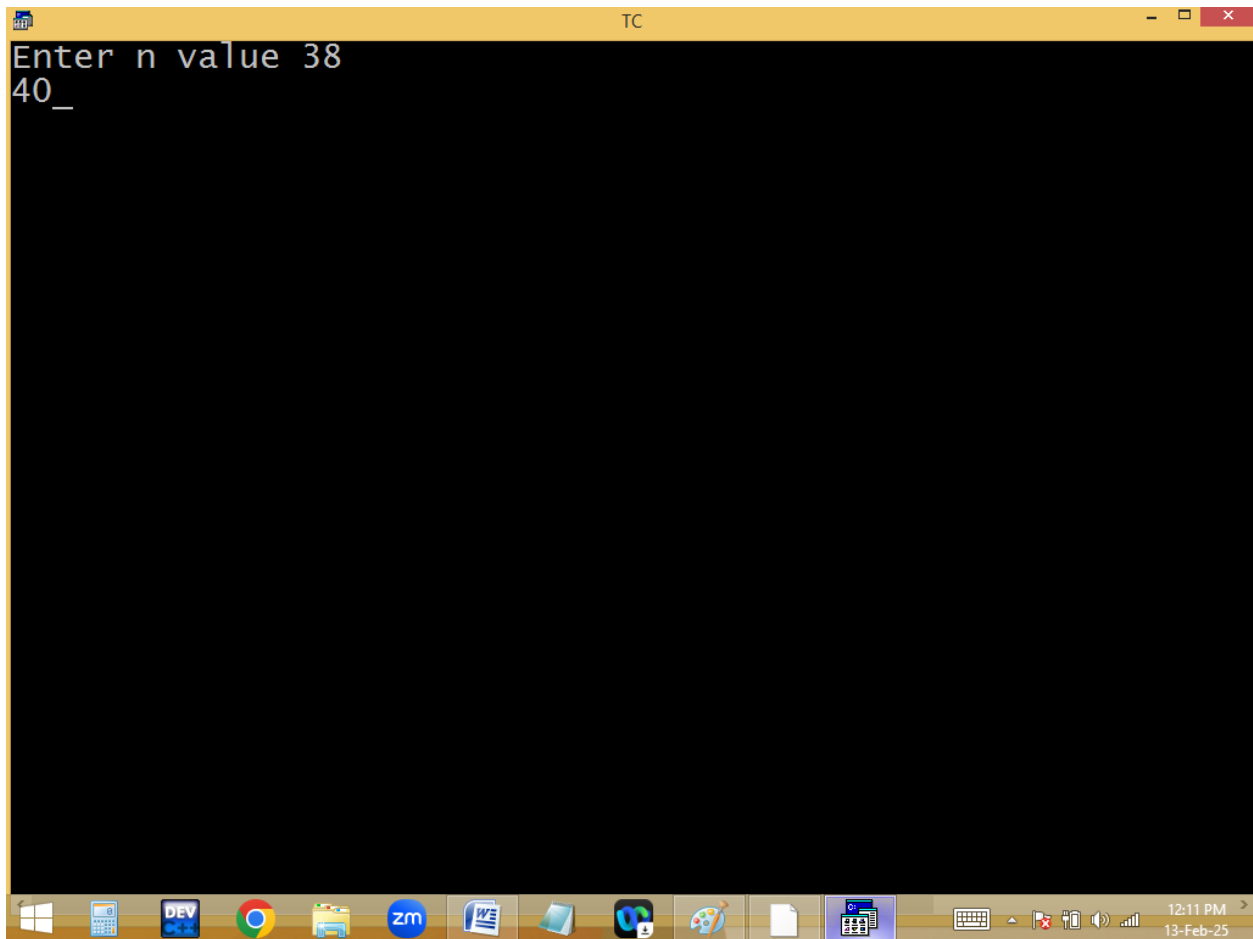


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 9", "Col 24", and a list of editing actions: "Insert", "Indent", "Tab", "Fill", "Unindent", and a cursor icon. The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
int n;
clrscr();
printf("Enter n value ");
scanf("%d",&n);
n%10>=5 && printf("%d",_n/10+1)*10)||printf("%d", n/10*10);
getch();
}
```

Below the code editor, there is a toolbar with function key shortcuts: "F1-Help", "F5-Zoom", "F6-Switch", "F7-Trace", "F8-Step", "F9-Make", and "F10-Run". The Windows taskbar is visible at the bottom, showing icons for the Start menu, Task View, File Explorer, Google Chrome, Zoom, and several other applications. The system clock in the bottom right corner indicates the time is 12:11 PM on 13-Feb-25.





```
TC
Enter n value 38
40_
```

Bitwise operators:

They works on bits [0,1]

It is one of the low level feature of c.

They works only int data.

C is 16 bit compiler. Hence the bitwise operations are conducted from 2^0 to 2^{15}

When the starting bit is 0, the given no is +Ve. If the starting bit is 1 the given no –Ve. Hence the starting bit is called sign bit.

Eg:

0111000 → +Ve

1011000 → -Ve

We have to consider only the 1 bits not 0 bits. Because of any no * 0 is 0.

They are used to develop os, device drivers and translators. They are faster than normal operators.

A – 97 - binary – processor – output

Eng - ascii – decimal no – binary

In c we are using 6 bitwise operators.

1. & - bitwise and

2. | - bitwise or

3. ^ - Xor – Exclusive or

4. ~ - compliment op

5. << - left shift op

6. >> - right shift op

1. & - when both bits are 1 then result also 1.
otherwise 0

$$25 \& 15 = 9$$